# The Safety Illusion: AI Signal Homogeneity and Systemic Financial Fragility

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**Abstract**

When financial institutions rely on the same AI models, their signals become correlated. We parameterise this homogeneity through a single correlation primitive $\rho$ and show that rising $\rho$ simultaneously activates three destabilising channels: it restores equilibrium multiplicity in a coordination game, collapses the information content of prices, and triggers correlated market-maker withdrawal. Because a common primitive drives all three, the channels amplify each other: the interior equilibrium loses stability at a compound threshold $\rho$ strictly below every individual channel threshold. Regulators examining any single channel in isolation will therefore find no cause for concern at correlation levels where the system has already crossed its compound fragility point. We call this the safety illusion. Competitive adoption endogenously drives $\rho$ past this threshold through a prisoner's dilemma; a diversity mandate corrects the externality.

## 1 Introduction

As artificial intelligence permeates financial markets, the signals it produces are converging. Diverse machine learning methods select the same dominant predictors of equity returns

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†This paper is a submission to the Human x AI Collaboration Challenge. AI has been used extensively in several steps of the research process.

[Gu et al., 2020], large language models at fixed temperature generate near-identical financial statement analyses [Kim et al., 2024] and autonomously converge on correlated market strategies [Fish et al., 2024], and market makers increasingly calibrate to common AI-driven models. This convergence is a structural feature of training on common data at scale, one that mirrors the factor-model crowding that precipitated the August 2007 quant crisis [Khandani and Lo, 2011, Brunnermeier, 2009].

The central question is not whether AI makes individual agents better informed, but what happens to financial stability when it makes them all informed in the same way.

We formalise AI signal homogeneity through a single primitive: the cross-sectional correlation $\rho \in [0, 1]$ of AI-generated signals. Each agent's signal error decomposes into a common AI model error shared by all AI-equipped agents and an idiosyncratic component, with $\rho$ governing the fraction of variance attributable to the common component. At $\rho = 0$, signals are independent and the economy sits at the Morris-Shin heterogeneous-signal benchmark. At $\rho = 1$, every AI-equipped agent observes the same signal, and the economy collapses to a common-prior world.

The core mechanism is that a single primitive drives three distinct channels of fragility. In the Goldstein and Pauzner [2005] coordination framework, rising $\rho$ converts heterogeneous private signals into a near-common signal, restoring the equilibrium multiplicity that Hellwig [2002] showed vanishes under sufficient signal heterogeneity; the crisis region expands non-monotonically in $\rho$. In the Grossman and Stiglitz [1980] information-aggregation framework, prices lose their capacity to reveal genuinely new information as the share of common signal variance grows, even as individual forecast accuracy improves. In the Avellaneda and Stoikov [2008] market-making framework, the effective number of independent liquidity providers collapses toward one. Each channel maps from the same object, $\rho$, to a different dimension of systemic stress.

Because $\rho$ activates all three channels simultaneously, the channels do not simply coexist; they amplify each other. Coordination failure makes prices less informative (agents act on correlated signals rather than revealing private information through trade), information loss raises the return to correlated strategies (reinforcing coordination failure), and both effects concentrate liquidity risk in common positions. We formalise this feedback through a composite operator and show that the compound fragility threshold satisfies

$$\rho^* < \min(\rho_1^*, \rho_2^*, \rho_3^*). \tag{1}$$

There exists a non-empty interval $(\rho^*, \min_i \rho_i^*)$ in which every single-channel stress test returns a favourable verdict while the system has already crossed its compound threshold.

We call this the *safety illusion*: at our baseline calibration, $\rho^* \approx 0.21$ sits well below each individual channel threshold, implying that regulators examining coordination risk, information quality, and liquidity provision in isolation will systematically underestimate systemic fragility.

The safety illusion would be a curiosity if $\rho$ were exogenous. It is not. Competitive AI adoption endogenously drives signal correlation above $\rho^*$ through a prisoner's dilemma: individual fund managers adopt AI to avoid tracking error relative to peers, but collective adoption pushes the system past the compound fragility threshold. A diversity mandate that caps effective signal correlation corrects this market failure.

Three recent papers study AI and financial stability. Danielsson and Uthemann [2025] model AI-driven fragility in a linear framework where common information shocks average out across agents; our non-linear specification preserves the correlation structure that causes thresholds to compound. Yang [2024] demonstrates computationally that reinforcement-learning agents converge on correlated strategies; our analytical framework characterises the coordination threshold explicitly as a function of signal correlation. Hansen and Lee [2025] find that independently queried AI agents reduce herding, consistent with our low-$\rho$ regime; we extend the analysis to high-$\rho$ environments where the herding effect dominates.

**Related literature.** The global games literature [Goldstein and Pauzner, 2005, Hellwig, 2002, Angeletos and Pavan, 2007] establishes that signal correlation structure determines equilibrium uniqueness and the welfare costs of correlated actions. Bebchuk and Goldstein [2011] apply global-games logic to credit market freezes driven by correlated withdrawal. These models treat signal correlation as a fixed parameter; we parameterise it continuously through $\rho$ and show that AI adoption endogenously restores the multiplicity that heterogeneous private signals were designed to eliminate.

The information acquisition literature, building on Grossman and Stiglitz [1980] and the forecasting-versus-revelatory distinction of Bond et al. [2012], studies how costly information shapes price efficiency.[1] Many agents processing the same AI output collapse aggregate information content even when each signal is individually precise; the information complementarity identified by Goldstein and Yang [2015] reverses as homogeneity rises.

On the supply side, our effective liquidity measure $N_{\text{eff}}(\rho)$ extends the demand-side fragility index of Greenwood and Thesmar [2011] to the Avellaneda and Stoikov [2008] market-making setting, with the $\rho = 1$ limit recovering the procyclical VaR spirals of Danielsson et al. [2012]. Brunnermeier [2009] provides the canonical empirical narrative

[1] Dugast and Foucault [2018] and Farboodi and Veldkamp [2020] analyse the temporal dimension of data growth and information acquisition; we address the cross-sectional dimension.

of this interaction (margin spirals, information loss, and correlated position unwinds during 2007–2008); our model formalises these dynamics through a single correlation primitive.

The broader AI-in-finance literature [Danielsson et al., 2022, Gu et al., 2020, Kleinberg and Raghavan, 2021] identifies the ingredients qualitatively, empirically, and through the lens of algorithmic monoculture; we supply a formal framework that unifies them with a compound fragility threshold as the new object.[2]

Section 2 introduces the signal structure. Sections 3.1–3.3 analyse coordination failure, information collapse, and liquidity withdrawal in isolation. Section 4 formalises the cross-channel feedback and derives the compound threshold. Section 5 endogenises $\rho$ through competitive adoption and characterises the corrective diversity mandate. Section 6 concludes.

# 2 Model Primitives

## 2.1 Signal Structure

We introduce the signal structure that underlies all three fragility channels. Each agent $i$ observes

$$x_i = \theta + \varepsilon_i, \tag{2}$$

where $\theta$ denotes the payoff-relevant fundamental and the noise term admits the factor decomposition

$$\varepsilon_i = \sqrt{\rho}\,\eta + \sqrt{1-\rho}\,\xi_i, \tag{3}$$

where $\eta \sim \mathcal{N}(0, \sigma^2)$ is a common error representing the correlated component introduced by shared training data, shared architecture, or a shared data vendor, and $\xi_i \sim \mathcal{N}(0, \sigma^2)$ is an idiosyncratic error capturing agent-specific model variation (distributional assumptions are stated formally in Assumption 1 below). The parameter $\rho$ measures the fraction of signal variance attributable to the common component.

*Assumption* 1 (Signal Errors). $\eta \sim \mathcal{N}(0, \sigma^2)$ and $\xi_i \sim \mathcal{N}(0, \sigma^2)$ are jointly normal, mutually independent, and independent across agents. Total signal variance $\mathrm{Var}(\varepsilon_i) = \sigma^2$ is therefore independent of $\rho$, so that comparative statics isolate the effect of redistributing noise between common and idiosyncratic components rather than changes in overall signal quality.

[2] Dou et al. [2025a] and Dou et al. [2025b] study AI-powered trading and market fragility in complementary settings; Barucca and Morone [2025] analyse how low-cost AI approximators reshape efficiency.

We write $\tau \equiv 1/\sigma^2$ for individual signal precision. The pairwise correlation follows directly:

$$\mathrm{Corr}(\varepsilon_i, \varepsilon_j) = \rho \quad \text{for } i \neq j. \tag{4}$$

At $\rho = 0$, signals are independent and the economy reduces to the heterogeneous-signal benchmark of Morris and Shin [1998]. At $\rho = 1$, every agent observes the same signal. Two distinct fundamentals appear across the channels – asset quality $\theta \sim \mathrm{Uniform}[0, 1]$ in the coordination channel, and a risky asset value $V$ in the information and liquidity channels – but the channels are unified by the common signal structure, not by a common fundamental. Each channel section specifies its own payoff-relevant state.

*Assumption* 2 (Equicorrelation). All pairs of AI-equipped agents share the same pairwise signal correlation $\rho$; the common factor $\eta$ is one-dimensional. Multi-factor extensions with heterogeneous $\rho_{ij}$ are an important direction for future work; see the limitations discussion in Section 6.

Equicorrelation is the worst case for aggregate action variance at a given mean pairwise correlation: by the Schur-convexity of the quadratic form $\mathbf{1}'\Sigma\mathbf{1}$, heterogeneous $\rho_{ij}$ with the same mean would raise the compound threshold $\rho^*$ but could introduce localised fragility in vendor-specific clusters.

## 2.2 Agents and Information Choices

The economy contains three types of agents: a fraction $\lambda \in (0, 1)$ of AI-equipped agents who observe signals with the $\rho$-parameterised structure in (3) at near-zero cost; a fraction $1 - \lambda$ of traditional agents who observe purely idiosyncratic signals ($\rho = 0$); and $N$ market makers who post competitive bid-ask quotes and manage inventory risk.[3]

An endogenous fraction $\mu_I \in [0, 1]$ of AI-equipped agents may also acquire a genuinely private signal $s_i^P = V + \zeta_i$, with $\zeta_i \sim \mathcal{N}(0, \sigma_P^2)$ independent of $\eta$, at cost $c_P > 0$. The choice between cheap but correlated AI signals and costly but independent private signals drives the information channel and connects to coordination and liquidity through the amplification loop.

The timing is as follows. At date 0, nature draws $\theta$; agents observe AI signals; AI-equipped agents decide whether to acquire private signals; agents choose actions. At date 1, payoffs realise based on $\theta$ and the aggregate action profile. Because $\rho$ enters through distinct strategic mechanisms, each channel requires its own equilibrium concept: a symmetric

[3] The primitive $\rho$ (intensive margin of signal homogeneity) is distinct from the $\mu$ parameter of Danielsson and Uthemann [2025] (extensive margin of AI penetration). Even low $\lambda$ with high $\rho$ can activate fragility, while high $\lambda$ with low $\rho$ may not.

threshold Bayesian Nash equilibrium in the coordination channel [Goldstein and Pauzner, 2005], a rational expectations equilibrium with endogenous information acquisition in the information channel [Grossman and Stiglitz, 1980], and a competitive equilibrium among liquidity providers in the liquidity channel. Each channel section defines its concept precisely; the amplification loop (Section 4) formalises how they compose.

### 2.3 Information Collapse

The signal structure implies a sharp bound on aggregate information that underpins the results in all three channels.

**Lemma 1** (Information Collapse)**.** *When $N$ agents each observe signals with pairwise correlation $\rho > 0$, their aggregate information content converges, as $N \to \infty$, to that of $N_{info}(\rho) = 1/\rho$ independent agents.*

*Proof.* The result follows from inverting the equicorrelation covariance matrix and computing the Fisher information limit as $N \to \infty$. See Appendix A. □

When $\rho = 1$, a thousand agents are informationally equivalent to one. When $\rho = 0.5$, to two. The common error $\eta$ cannot be diversified away by adding agents, because all agents err in the same direction. This has an immediate consequence for coordination: $\eta$ functions as an endogenous public signal that can trigger collectively self-fulfilling actions at correlation levels well below what individual signal quality would suggest.

## 3 Three Fragility Channels

### 3.1 Coordination Failure

Rising AI signal correlation does not uniformly destabilise. At low $\rho$, correlation actually lowers the crisis threshold, a "wisdom of AI" effect in which better-informed agents coordinate more efficiently. Only above a tipping point $\rho_0$ does herding take over, driving the threshold upward until equilibrium uniqueness collapses entirely at $\rho_1^*$. This non-monotonicity, absent from linear frameworks such as Danielsson and Uthemann [2025], is the section's central result; it arises from the interaction between payoff asymmetry and higher-order beliefs about the common shock, and it disappears when the payoff from staying equals the loss from a failed bank.

### 3.1.1 The Bank-Run Game

For the standalone analysis all depositors are AI-equipped ($\lambda = 1$); the heterogeneous-agent case enters through the amplification loop of Section 4. We seek a symmetric monotone Bayesian Nash equilibrium in threshold strategies: each agent withdraws if and only if $x_i < x^*$.

The departure from Goldstein and Pauzner [2005] is immediate. Because AI signals share the common shock $\eta$, the aggregate withdrawal fraction depends on its realisation:

$$\ell_{\mathrm{AI}}(\theta, \eta, x^*) = \Phi\left(\frac{x^* - \theta - \sqrt{\rho}\,\eta}{\sqrt{1-\rho}\,\sigma}\right), \tag{5}$$

where $\Phi$ is the standard normal CDF. In Goldstein and Pauzner [2005], idiosyncratic errors wash out in the aggregate and $\ell$ depends on $\theta$ alone. Here, a single adverse realisation of $\eta$ can push $\ell$ above $\theta$ even when fundamentals are sound; the common shock creates correlated withdrawal that fundamentals-based stress tests would not predict. The crisis threshold $\theta^*(\rho)$ is defined by

$$\mathbb{E}_\eta[\ell(\theta^*, \eta, x^*)] = \theta^*. \tag{6}$$

### 3.1.2 Uniqueness and Its Breakdown

The common shock $\sqrt{\rho}\,\eta$ shifts all signals identically, functioning as a public signal in the sense of Hellwig [2002]. Rewriting the signal as $x_i = \tilde{\theta} + \sqrt{1-\rho}\,\xi_i$, where $\tilde{\theta} \equiv \theta + \sqrt{\rho}\,\eta$ is the effective fundamental, reduces the problem to a standard global game conditional on a realisation of $\eta$, but with the effective fundamental now random even conditional on $\theta$. Applying Hellwig's contraction condition to this structure yields:

**Proposition 1** (Uniqueness-Multiplicity Boundary)**.** *The threshold equilibrium is unique if and only if $\rho < \rho_1^*$, where*

$$\rho_1^* = \frac{1}{1 + \alpha_{SC}^2} \tag{7}$$

*and $\alpha_{SC} = R(\theta^*)/(R(\theta^*) + 1 - \theta^*) \in (0, 1)$ is the strategic complementarity parameter. For $\rho > \rho_1^*$, the global game admits multiple equilibria.*

*Proof sketch.* The Goldstein-Pauzner payoff structure satisfies monotone best responses, dominance regions, and normal signals. The contraction condition requires $\alpha_{\mathrm{SC}} \cdot \sqrt{\rho/(1-\rho)} < 1$, giving (7). Full verification in Appendix D. □

The uniqueness boundary $\rho_1^*$ complements the analysis of Bebchuk and Goldstein [2011], who study self-fulfilling credit market freezes in a global-games framework with correlated

fundamentals. Their model generates multiplicity through common exposure to borrower quality; our parameterisation through $\rho$ additionally captures the case where correlation arises endogenously from shared AI signals rather than shared fundamentals, and integrates the coordination channel with information acquisition and liquidity provision through the compound threshold.

### 3.1.3 Non-Monotonicity of the Crisis Threshold

The crisis threshold does not rise uniformly with signal correlation. This is the section's central result, and it distinguishes the present analysis from the linear threshold of Danielsson and Uthemann [2025].

**Proposition 2** (Crisis Threshold Non-Monotonicity)**.** *For $\rho \in (0, \rho_1^*)$, the unique crisis threshold $\theta^*(\rho)$ is non-monotone: there exists $\rho_0 \in (0, \rho_1^*)$ such that*

$$\frac{d\theta^*}{d\rho} < 0 \quad \textit{for } \rho \in (0, \rho_0), \qquad \frac{d\theta^*}{d\rho} > 0 \quad \textit{for } \rho \in (\rho_0, \rho_1^*). \tag{8}$$

*The payoff asymmetry $R(\theta) > 1$ is necessary for this non-monotonicity.*[4]

*Proof sketch.* The aggregate withdrawal condition integrates over $\eta$ via the probit convolution identity, so non-monotonicity operates through the individual indifference condition. At low $\rho$, idiosyncratic uncertainty dominates; at high $\rho$, common uncertainty dominates. The sign reversal follows from the interaction of payoff asymmetry with the changing composition of posterior uncertainty. Full derivation in Appendix D. □

The economics of the sign reversal (Figure 1) are rooted in higher-order beliefs. At low $\rho$, a depositor who receives a bad signal faces genuine uncertainty about whether others received bad signals too: idiosyncratic noise dominates, so one bad draw does not reliably predict a coordinated run. Since the payoff from staying exceeds the loss from a failed bank ($R(\theta) > 1$), this uncertainty creates an option value of staying that lowers $\theta^*$. At high $\rho$, noise is mostly common, so a bad signal reliably predicts that all other AI-equipped depositors received similarly bad signals. The depositor knows a run is coming, and withdrawal becomes individually optimal regardless of fundamentals. The SVB episode of March 2023, where depositors sharing common beliefs about interest-rate risk coordinated a \$42 billion withdrawal in a single day [Jiang et al., 2023], illustrates the high-$\rho$ regime at the extreme.

This non-monotonicity is absent from the Danielsson and Uthemann [2025] model. Their aggregate withdrawal condition integrates out the common shock before determining the

[4] The sufficient condition on signal noise is $\sigma < \sqrt{1-\rho_1^*}/(2\sqrt{2\pi})$, ensuring the posterior concentrates in the dominance region interior. Weaker conditions are given in Appendix D.

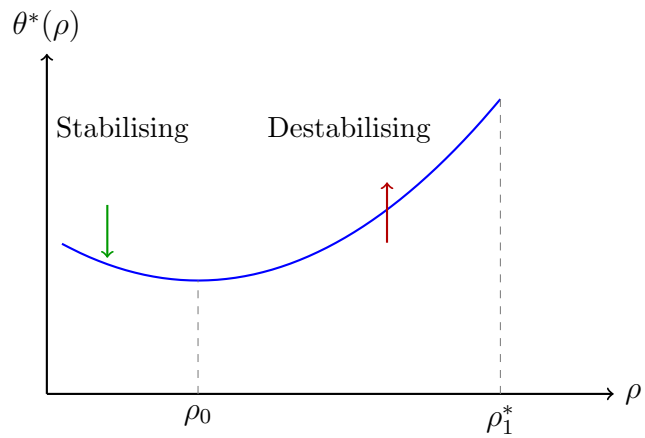


Figure 1: Crisis threshold $\theta^*(\rho)$ for $\rho \in [0, \rho_1^*)$, where $\rho_1^* = 1/(1 + \alpha_{\text{SC}}^2)$ is the uniqueness-multiplicity boundary. For $\rho < \rho_0$, rising AI signal correlation lowers the crisis threshold (stabilising regime): better-correlated signals improve coordination efficiency. For $\rho > \rho_0$, herding dominates and the threshold rises sharply (destabilising regime). At $\rho_1^*$, equilibrium uniqueness collapses and multiple self-fulfilling outcomes coexist. The compound fragility threshold $\rho^* \approx 0.21$ lies below $\rho_1^*$, so the system enters the fragile region well before uniqueness breaks down. Parameters: $\sigma = 0.05$, $R = 1.5$, $\lambda = 1$.

threshold, producing the linear relationship $\theta^*(\mu) = (c/b)[(1-\mu)p + \mu]$. Preserving $\eta$ in the withdrawal fraction is what allows the interaction between payoff asymmetry and higher-order beliefs to generate the U-shape.[5]

#### 3.1.4 Welfare

In the Angeletos and Pavan [2007] linear-quadratic approximation, the social welfare loss from signal correlation is

$$W_{\text{loss}}(\rho) = \frac{\alpha_{\text{SC}}^2}{(1-\alpha_{\text{SC}})^2} \cdot \lambda^2 \cdot \frac{\rho}{\tau}, \tag{9}$$

linearly increasing in $\rho$, while the private benefit of correlation is decreasing: $V_{\text{priv}} = (1-\rho)/\tau$. The socially optimal correlation $\rho^{\text{SO}}$ therefore lies strictly below $\rho_1^*$ for any $\alpha_{\text{SC}} > 0$. The Angeletos-Pavan formula is a linear-quadratic benchmark for a binary-action game, so it captures the qualitative direction of the welfare cost rather than its precise magnitude. Full derivation and the binary-game approximation caveat are in Appendix D.

The coordination analysis establishes that AI signal homogeneity creates fragility through an information-theoretic mechanism: the common shock functions as an endogenous public signal that raises the crisis threshold and ultimately destroys equilibrium uniqueness. But

[5] The comparative statics are also qualitatively distinct: Danielsson and Uthemann [2025] parameterise the extensive margin (fraction of AI agents $\mu$), while we parameterise the intensive margin (signal correlation $\rho$). Even a small fraction of AI agents can destabilise if their signals are sufficiently correlated.

this does not explain why the market's capacity to aggregate information simultaneously deteriorates. Even within the uniqueness region, agents acting on increasingly similar signals erode the diversity that makes prices informative, a degradation that proceeds silently, invisible to standard forecast-accuracy metrics.

## 3.2 Information Acquisition Collapse

Standard forecast-accuracy metrics suggest that AI adoption improves price efficiency. The opposite is true: revelatory price efficiency (the information that prices convey beyond the common AI signal) declines monotonically in $\rho$, even as forecasting efficiency improves. The divergence renders the deterioration invisible to conventional measurement.

### 3.2.1 Information Acquisition in a Grossman-Stiglitz Framework

We embed the signal structure into the Grossman and Stiglitz [1980] CARA-normal rational expectations framework with noise trader demand $u \sim \mathcal{N}(0, \sigma_u^2)$. The fundamental is $V \sim \mathcal{N}(\bar{v}, 1/\tau_v)$, distinct from the uniform $\theta$ of the coordination channel; the channels are unified by the signal correlation structure (3), not by a common fundamental. Agents are price-takers in the Verrecchia [1982] competitive equilibrium and choose among three information states: *AI-informed* (near-zero cost, signals with pairwise correlation $\rho$), *privately informed* (cost $c_P > 0$, independent signals with precision $\tau_P = 1/\sigma_P^2$), and *uninformed* (observing only the price). Let $\mu_I$ denote the endogenous fraction acquiring private information. By the information collapse lemma, the aggregate AI signal reveals $V + \sqrt{\rho}\,\eta$ with effective precision $1/(\rho\sigma_s^2)$ regardless of how many AI agents participate; the price ceases to reveal information beyond the common AI component.

### 3.2.2 Trading Rents and Information Fractions

The CARA-normal equilibrium [Verrecchia, 1982, Back et al., 2000] yields AI trading rents

$$\pi_A(\rho, \mu_A) \approx k_A \cdot (1-\rho)^{\alpha} \cdot \mu_A^{-\beta}, \tag{10}$$

where $k_A = \tau/(2\gamma\sigma_u)$, and $\alpha \approx 0.04$, $\beta \approx 0.58$ are obtained by numerical calibration of the full REE solution at the baseline parameters in Table 1. The exponent on $(1-\rho)$ is the key: $\alpha \approx 0.04$ means AI rents decline by only approximately 13% as $\rho$ rises from 0 to 0.99. The $\rho$-dependence enters entirely through the price informativeness channel, but noise trader demand attenuates the effect; when the price is already noisy from non-informational

trading, the additional noise from correlated AI errors is a small perturbation. For privately informed agents, the standard result applies: $\pi_P(\mu_I) = k_P/\mu_I$ where $k_P = \tau_P/(2\gamma\sigma_u)$.

This near-invariance has a structural consequence. Crisis risk from the coordination channel raises the effective cost of private information, because the informed position has value only when the institution survives:

$$\pi_P(\mu_I^*) - \pi_A(\rho, 1-\mu_I^*) = \frac{c_P}{1-\theta^*(\rho)}. \tag{11}$$

As the crisis threshold $\theta^*$ rises, the denominator shrinks, making private information effectively more expensive.[6] One might expect agents to substitute toward private signals as AI rents erode, partially restoring information diversity. The next result shows this self-correction fails.

**Lemma 2** (Information Diversity Collapse)**.** *In the extended Grossman and Stiglitz [1980] model with AI signals (cost $c_{AI} = 0$, pairwise correlation $\rho$) and private signals (cost $c_P > 0$, correlation 0):*

(i) *Within the information channel alone, $\mu_I^*(\rho)$ is weakly increasing in $\rho$: declining AI rents make private information relatively more attractive.*

(ii) *When crisis risk is incorporated via* (11)*, the net effect reverses above a threshold: $d\mu_I^*/d\rho < 0$, because crisis risk dominates the within-channel substitution.*

(iii) *Regardless of the direction of $\mu_I^*(\rho)$, the effective number of independent information sources*

$$N_{info}(\rho, \mu_I) = \mu_I N + \frac{(1-\mu_I)N}{1 + ((1-\mu_I)N - 1)\rho} \tag{12}$$

*is strictly decreasing in $\rho$ for any fixed $\mu_I$.*

*Proof sketch.* Part (i): Weak $\rho$-dependence of AI rents ($\alpha \approx 0.04$) induces modest substitution toward private information. Part (ii): The crisis-risk term $c_P/(1-\theta^*)$ dominates in the fragile region. Part (iii): Direct computation from the equicorrelation structure. Full derivation in Appendix E. □

Part (iii) is the structural result. The competing forces in parts (i) and (ii), substitution toward private signals versus crisis risk destroying their option value, are second-order relative to the mechanical collapse of $N_{\text{info}}$ driven by equicorrelation itself. No amount of substitution toward private signals can offset the redundancy that correlation introduces among AI signals. The cross-sectional dimension of information loss, operating even when

[6] In the standalone analysis, $\theta^*(\rho)$ uses exogenous $\rho$. In the amplification loop (Section 4), this generalises to $\theta^*(\rho_{\text{eff}})$ with $\rho_{\text{eff}} \geq \rho$ incorporating cross-channel feedback.

each individual signal remains precise, is distinct from the temporal dimension studied by Dugast and Foucault [2018] and Farboodi and Veldkamp [2020].

### 3.2.3 The FPE-RPE Divergence

**Proposition 3** (Price Informativeness)**.** (i) *Forecasting price efficiency* FPE($\rho$)*, measuring how well the price predicts $V$, is non-monotone in $\rho$: increasing for $\rho < \rho^*_{FPE}$ and decreasing thereafter.*

(ii) *Revelatory price efficiency*

$$\mathrm{RPE}(\rho) = \frac{[\mu_I^*(\rho)]^2 \tau_P}{\gamma^2 \sigma_u^2} + \frac{(1-\mu_I^*(\rho))(1-\rho)\tau}{\gamma^2 \sigma_u^2}, \tag{13}$$

*measuring the novel information the price reveals beyond the common AI signal, is monotonically decreasing in $\rho$ under the sufficient condition $\mu_I^*(0) < \tau/(2\tau_P)$, which holds at the baseline calibration for $c_P > 0$.*

(iii) *For $\rho \in (0, \rho^*_{FPE})$, FPE improves while RPE declines.*

*Proof sketch.* FPE is non-monotone because information collapse eventually outweighs initial diversity gains from cheap AI signals. RPE is monotonically decreasing because direct erosion of novel information dominates the composition effect. The divergence follows from their definitions: FPE includes the common AI component; RPE does not. Full derivation in Appendix E. □

The divergence in part (iii) is the policy-relevant result. A regulator measuring forecast accuracy for $\rho < \rho^*_{\mathrm{FPE}}$ observes improving performance and concludes that AI adoption enhances price quality. This inference fails: the price simultaneously reveals less that is genuinely new. A firm manager observing the market price learns about the common AI assessment, which she could obtain herself at near-zero cost. She learns nothing new from the price.

The information loss extends beyond depth to breadth. Goldstein and Yang [2015] show that information diversity creates strategic complementarities in acquisition: knowing one dimension of a firm's fundamentals makes investigating other dimensions more valuable. When AI homogenises one dimension, this complementarity measure $C_{\mathrm{GY}}(\rho)$ declines strictly in $\rho$ and approaches zero as $\rho \to 1$ (derivation in Appendix I). A market dominated by AI signals that agree on one dimension loses the incentive structure that sustains investigation of other dimensions.

Informationally impoverished prices need not be fatal if liquidity remains robust. Whether it does depends on the market makers who absorb order flow, and when those market makers

share AI-calibrated models, their quoting decisions become correlated for the same structural reason that trading signals do.

## 3.3 Correlated Liquidity Exit

At $N = 100$ market makers and $\rho = 0.1$, the effective number of independent liquidity providers is $N_{\text{eff}} \approx 28$. At $\rho = 0.3$, it falls below 10. This convex collapse, where most diversification is lost at low correlation, means that even modest AI calibration similarity dramatically concentrates liquidity supply, creating conditions for simultaneous exit well before individual market makers recognise the systemic exposure.

### 3.3.1 Market Making with Correlated Calibrations

Consider $N$ market makers, each using an AI-calibrated model to post bid-ask quotes in a single asset with value $V \sim \mathcal{N}(\bar{v}, \sigma_V^2)$. In the Avellaneda and Stoikov [2008] stochastic control framework, each market maker $i$ posts a reservation price

$$r_i(t) = S_i(t) - q_i \gamma_i \sigma_i^2 (T - t), \tag{14}$$

where $S_i(t)$ is the midpoint estimate, $q_i$ is inventory, $\gamma_i$ is risk aversion, and $\sigma_i^2$ is the estimated asset variance. When market makers share AI-calibrated models, their estimates inherit the $\rho$-parameterised correlation structure:

$$\sigma_i^2 = \sigma_V^2 + \varepsilon_i^\sigma, \quad \varepsilon_i^\sigma = \sqrt{\rho}\,\eta_\sigma + \sqrt{1-\rho}\,\xi_i^\sigma, \tag{15}$$

where $\eta_\sigma \sim \mathcal{N}(0, \sigma_{\text{cal}}^2)$ is the common AI calibration error and $\xi_i^\sigma \sim \mathcal{N}(0, \sigma_{\text{cal}}^2)$ is idiosyncratic. A negative common error simultaneously raises all market makers' variance estimates, pushing them toward the exit threshold. Similarly, midpoint estimates $S_i(t) = V + \sqrt{\rho}\,\eta_S + \sqrt{1-\rho}\,\xi_i^S$ share the same correlation structure. Market maker $i$ exits (sets spread to infinity) when estimated variance exceeds a tolerance threshold $\sigma_{\max}^2$.

### 3.3.2 Effective Liquidity Diversification

Each market maker's exit decision is a Bernoulli random variable $W_i = \mathbf{1}\{\hat{\sigma}_i^2 > \sigma_{\max}^2\}$ with mean $p_w = \Pr[\hat{\sigma}_i^2 > \sigma_{\max}^2]$. Because the calibration errors are equicorrelated normals, the withdrawal indicators are not equicorrelated Bernoullis with pairwise correlation $\rho$. The

bivariate normal CDF gives the correct withdrawal correlation. Letting $c = \Phi^{-1}(1 - p_w)$:

$$\rho_W(\rho, p_w) = \frac{\Phi_2(-c, -c;\, \rho) - p_w^2}{p_w(1 - p_w)}, \tag{16}$$

where $\Phi_2(\cdot, \cdot;\, \rho)$ is the bivariate standard normal CDF. The withdrawal correlation satisfies $\rho_W(0, p_w) = 0$, $\rho_W(1, p_w) = 1$, and $\rho_W(\rho, p_w) \leq \rho$ for $p_w \neq 0.5$, with attenuation largest for tail events: at $\rho = 0.1$ and $p_w = 0.05$, $\rho_W \approx 0.028$, only 28% of the underlying Gaussian correlation. Equating the variance of the average withdrawal rate under correlated and independent providers yields the effective number of independent liquidity providers:

$$N_{\text{eff}}(\rho, p_w) = \frac{N}{1 + (N - 1)\rho_W(\rho, p_w)}. \tag{17}$$

**Lemma 3** (Liquidity Fragility)**.** *The effective number of independent liquidity providers $N_{\text{eff}}(\rho, p_w)$ satisfies:*

(i) $N_{\text{eff}}(0, p_w) = N$ *and* $N_{\text{eff}}(1, p_w) = 1$.

(ii) $N_{\text{eff}}$ *is strictly decreasing in* $\rho$.

(iii) $N_{\text{eff}}$ *is strictly convex in* $\rho$*: the marginal loss of independence is greatest at low correlation.*

*Proof sketch.* Parts follow from monotonicity and convexity of $\rho_W$ in $\rho$, verified from bivariate normal CDF properties. Full verification in Appendix K. □

The convexity in part (iii) is the quantitatively important result. At $N = 100$, $\rho = 0.1$, and $p_w = 0.05$, the corrected $N_{\text{eff}} \approx 28.4$, consistent with the opening claim that a hundred nominal market makers may behave as fewer than thirty. Detailed comparisons across parameter values are in the Internet Appendix. The $N_{\text{eff}}(\rho)$ formula extends the Greenwood and Thesmar [2011] stock fragility measure from the demand side to the supply side: where Greenwood and Thesmar capture how correlated mutual fund flows create instability in asset prices, the present measure captures how correlated AI calibrations create instability in liquidity supply. The mechanism is algorithmic similarity rather than ownership overlap. The May 2010 Flash Crash, in which algorithmic market makers sharing similar volatility models pulled quotes within minutes [Kirilenko et al., 2017], illustrates this collapse of $N_{\text{eff}}$ toward one.

The $N_{\text{eff}}$ measure is a structural primitive, not an empirical measurement tool. When $N_{\text{eff}}$ falls, the model predicts spread widening and correlated liquidity exit, both captured empirically by a rise in CoVaR [Adrian and Brunnermeier, 2016] and SRISK [Brownlees and Engle, 2017]. The present measure explains why these empirical risk indicators rise during

periods of AI-driven correlation: the effective number of independent liquidity providers shrinks, concentrating systemic exposure. This structural role, providing a micro-founded explanation for movements in reduced-form risk measures, is analogous to the function that the intermediary capital ratio of He and Krishnamurthy [2013] serves relative to empirical asset pricing factors.

### 3.3.3 The No-Equilibrium Threshold

Revenue per market maker is $Qs^*/N$, where $s^*$ is the equilibrium half-spread. Since $s^* = s_0/N_{\text{eff}}$, revenue scales linearly with the variance inflation factor $z = 1+(N-1)\rho_W$. Inventory cost, by contrast, scales quadratically: a correlated portfolio of $N$ positions has variance proportional to $z^2$, since margin requirements and risk charges typically scale with position variance, not position size [Brunnermeier and Pedersen, 2009]. Net profit per market maker is therefore $D \cdot z - C \cdot z^2 - \pi_0$, where $D = Qs_0/N^2$ is the revenue coefficient, $C = \gamma\sigma_V^2\kappa_{\text{inv}}$ is the quadratic inventory cost coefficient, and $\pi_0 > 0$ is the outside option.

**Proposition 4** (No-Equilibrium Threshold)**.** *Under regularity conditions ensuring the participation constraint binds (stated in Appendix K), there exists $\rho^{**} \in (0,1)$ such that:*

(i) *For $\rho < \rho^{**}$: net profit is positive and an interior market-making equilibrium with finite spreads exists.*

(ii) *For $\rho > \rho^{**}$: net profit is negative, all market makers withdraw simultaneously, and the market enters the Pagano [1989] low-liquidity trap.*

*The threshold is the upper root of the participation constraint:*

$$\rho^{**} = \frac{1}{N-1}\left(\frac{D + \sqrt{D^2 - 4C\pi_0}}{2C} - 1\right). \tag{18}$$

*Proof sketch.* Net profit $D \cdot z - C \cdot z^2 - \pi_0$ is a downward-opening quadratic in $z$ with two roots. The upper root gives $\rho^{**}$. For $\rho > \rho^{**}$, inventory cost dominates and market making is unprofitable. Full derivation in Appendix K. □

The simultaneous exit of all identically positioned market makers validates the high-volatility estimate that triggered the exit, completing a self-fulfilling liquidity collapse. At $\rho = 1$, all calibration errors are identical, so all $N$ market makers act as a single representative agent with $N_{\text{eff}} = 1$ and $s^* = s_0$. This limit recovers the Danielsson et al. [2012] common VaR representation, though the mechanisms differ: Danielsson, Shin, and Zigrand operate through regulatory VaR constraints and endogenous price impact, while the present channel

operates through correlated AI calibration errors.[7]

Correlated calibration in isolation understates the danger. When $N_{\text{eff}}$ falls, spreads widen, making it costlier to trade on private information, which reduces $\mu_I^*$ and concentrates reliance on AI signals. The effective signal correlation in the coordination game rises, pushing the economy closer to the uniqueness boundary. A higher crisis probability in turn further destroys the option value of private information. The compound effect of these interactions is the subject of the next section.

# 4 Cross-Channel Amplification

## 4.1 The Composite Operator

The three channels form a circular feedback system whose compound threshold lies strictly below every individual channel threshold. We formalise the joint dynamics as a fixed-point problem on the state vector $v = (\rho_{\text{eff}}, \theta^*, N_{\text{eff}}) \in K$, where

$$K = [\rho, 1] \times [\theta_L, \theta_H] \times [1, N] \tag{19}$$

is compact and convex. Three equilibrium mappings connect the state variables in a cycle: liquidity concentration raises effective correlation, higher effective correlation raises the crisis threshold, elevated crisis risk reduces private information acquisition, and the resulting AI reliance reduces liquidity diversity.

**Liquidity concentration amplifies effective correlation.** When the number of independent liquidity providers falls, the remaining market makers' correlated signals raise the effective correlation in the coordination game. The $g_1$ mapping captures this link:

$$\rho_{\text{eff}} = g_1(\rho, N_{\text{eff}}) = 1 - (1 - \rho)\frac{N_{\text{eff}}}{N}. \tag{20}$$

The linear form interpolates between $\rho$ (when $N_{\text{eff}} = N$) and 1 (when $N_{\text{eff}} = 1$); any monotone alternative satisfying these boundary conditions yields qualitatively identical results. The linear form is not microfounded; it is chosen for analytical tractability, and the quantitative implications are bracketed by the sensitivity analysis below. A concave $g_1$ is the natural specification if the most stabilising liquidity providers (informed dealers with large idiosyncratic signal components) exit first as concentration rises, so that the initial losses

[7] The $N_{\text{eff}}$ formula provides a continuous parametric path from the heterogeneous-calibration case to the DSZ $\rho = 1$ limit, without requiring the VaR regulatory apparatus.

of diversity have disproportionate impact on effective correlation. A convex $g_1$ arises under the opposite ordering: if redundancy among market makers provides diminishing marginal protection, each additional exit matters less until the last few providers remain. The linear baseline is agnostic between these mechanisms and sits near the centre of the robustness interval reported in the Internet Appendix.[8]

**Crisis risk reduces private information acquisition.** As the crisis threshold rises, the expected payoff from holding a privately informed position falls: signals have lower marginal value when coordination risk dominates. The $g_2$ mapping formalises this through the Grossman-Stiglitz indifference condition (11):

$$\mu_I^* = g_2(\theta^*, \rho_{\text{eff}}, c_P): \quad \pi_P(\mu_I^*) - \pi_A(\rho_{\text{eff}}, 1-\mu_I^*) = \frac{c_P}{1-\theta^*}. \tag{21}$$

**AI reliance reduces liquidity diversity.** With $N_{\text{eff}}$ independent market makers rather than $N$, liquidity provision becomes concentrated. The $g_3$ mapping connects this concentration back to effective correlation:

$$N_{\text{eff}} = g_3(\mu_I, \rho, N) = \frac{N}{1+(N-1)\rho(1-\mu_I)^2}. \tag{22}$$

The $g_3$ mapping uses the signal-level correlation $\rho$ rather than the withdrawal correlation $\rho_W$ from (16). The distinction matters because $g_3$ aggregates the variance of market makers' signal portfolios (the mix of AI and private information driving quote-setting), not the binary withdrawal indicators whose correlation $\rho_W$ governs Channel 3 in isolation. In the standalone Channel 3 analysis, $\rho_W$ is the correct object because the outcome of interest is the number of simultaneous exits, which are Bernoulli random variables with bivariate-normal-induced correlation. In the amplification loop, the outcome of interest is how concentrated the information content of surviving quotes becomes as AI reliance rises, which depends on signal correlation directly. Since $\rho_W(\rho, p_w) \leq \rho$ for all $p_w \neq 0.5$ (with the gap widest for tail withdrawal probabilities), using $\rho$ in $g_3$ is the conservative specification: substituting $\rho_W$ would raise $N_{\text{eff}}$ and therefore raise $\rho^*$, widening the safety illusion interval (Table IA.3 confirms $\rho^*$ rises from 0.215 to 0.380 under this substitution).

[8] The qualitative result $\rho^* < \min_i \rho_i^*$ holds for any monotone $g_1$ satisfying $g_1(\rho, N) = \rho$ and $g_1(\rho, 1) = 1$; see the Internet Appendix for robustness across alternative specifications.

## 4.2 Fixed-Point Existence

The composite operator $T : K \to K$ maps $v = (\rho_{\text{eff}}, \theta^*, N_{\text{eff}})$ through:

$$\begin{aligned} T_3(v) &= g_3(g_2(\theta^*, \rho_{\text{eff}}, c_P),\ \rho,\ N)\,, \\ T_1(v) &= g_1(\rho, T_3(v)) = 1 - (1-\rho)\frac{T_3(v)}{N}, \\ T_2(v) &= \theta^*(T_1(v)), \end{aligned} \tag{23}$$

where $\theta^*(\cdot)$ is the crisis threshold function from Proposition 2.

**Proposition 5** (Fixed-Point Existence)**.** *The composite operator $T : K \to K$ is continuous and maps the compact convex set $K$ into itself. By Brouwer's fixed-point theorem, $T$ admits at least one fixed point $(\rho^*_{\text{eff}}, \theta^{**}, N^*_{\text{eff}}) \in K$.*

*Proof.* $K$ is compact and convex. $T$ maps $K$ into $K$: $T_3(v) \in [1, N]$ because $\mu_I \in [0, 1]$; $T_1(v) \in [\rho, 1]$ because $T_3(v) \in [1, N]$; $T_2(v) \in [\theta_L, \theta_H]$ by the dominance argument. Each component ($g_1$ linear, $g_3$ rational, $g_2$ and $\theta^*$ $C^1$ by the implicit function theorem) is continuous, so $T$ is continuous. Brouwer's theorem applies. □

## 4.3 Feedback Strength and the Dominant Eigenvalue

At an interior fixed point, the feedback loop strength is governed by two chains of partial derivatives (Appendix A, equations (28)–(33)). The crisis-sensitivity chain captures how liquidity loss propagates through the coordination game: $h \equiv d\theta^*/d\rho_{\text{eff}} > 0$ (higher effective correlation raises the crisis threshold) and $|a| \equiv |d\mu^*_I/d\theta^*|$ (elevated crisis risk reduces private research). The liquidity-concentration chain captures how reduced private research feeds back into liquidity: $w \equiv (1-\rho)/N > 0$ (the $g_1$ amplification strength) and $m \equiv dg_3/d\mu_I > 0$ (fewer private researchers raise market-maker AI reliance). Working against the loop is the stabilising force $b \equiv d\mu^*_I/d\rho_{\text{eff}} > 0$: higher correlation makes private signals relatively more attractive, partially offsetting the crisis effect.

The Jacobian $DT$ has rank two. Its dominant eigenvalue is

$$\lambda_+ = \frac{wm}{2}\left[(h|a| - b) + \sqrt{(h|a| - b)^2 + 8h|a|b}\right]. \tag{24}$$

The discriminant is strictly positive, so $\lambda_+$ is real. When $\lambda_+$ crosses unity, the interior fixed point loses stability and the feedback loop amplifies perturbations rather than dampening them.

## 4.4 Bifurcation and the Safety Illusion

We first state the equilibrium selection assumption under which the compound fragility results operate.

*Assumption* 3 (Interior Equilibrium Selection). The economy operates at an interior fixed point of the composite operator $T$: that is, a fixed point $v^* = (\rho^*_{\text{eff}}, \theta^{**}, N^*_{\text{eff}}) \in K$ with $\rho^*_{\text{eff}} < \rho^*_1$, $\mu^*_I > 0$, and $N^*_{\text{eff}} > 1$.

Assumption 3 selects the economically relevant equilibrium in which private information acquisition is active and multiple independent liquidity providers operate. For $\rho < \rho^*$, the interior fixed point is locally stable ($\lambda_+ < 1$) and locally unique by the Ostrowski theorem (Appendix F), so the selection is natural. The assumption is substantive for $\rho > \rho^*$, where the interior fixed point coexists with corner equilibria (Remark F).

**Proposition 6** (Loss of Interior Stability). *Let $\rho^*_1$, $\rho^*_2$, $\rho^*_3$ denote the individual channel thresholds from Propositions 1, 3, and 4. Under Assumptions 3 and 4, define $\rho^*$ as the exogenous correlation at which the spectral radius of $DT$ evaluated at the interior fixed point crosses unity. Then*

$$\rho^* < \min(\rho^*_1, \rho^*_2, \rho^*_3). \tag{25}$$

*Proof sketch.* The crisis sensitivity $h = d\theta^*/d\rho_{\text{eff}}$ diverges as $\rho_{\text{eff}} \to \rho^{*-}_1$ via the saddle-node expansion in Appendix C (which requires Assumption 4). As $h \to \infty$, $\lambda_+ \sim wm\sqrt{h|a|b} \to \infty$, so the crossing $\lambda_+ = 1$ occurs at finite $\rho_{\text{eff}} < \rho^*_1$. Since $g_1$ ensures $\rho^*_{\text{eff}} > \rho$ whenever $N^*_{\text{eff}} < N$, the exogenous compound threshold satisfies $\rho^* < \rho^*_1$. Analogous arguments for Channels 2 and 3 complete the strict inequality. Full proof in Appendix H. □

The inequality (25) implies a non-empty interval $(\rho^*, \min_i \rho^*_i)$ in which every individual channel assessment is favourable but the interior fixed point of the integrated system has lost stability.

**Corollary 1** (Safety Illusion). *Suppose Assumption 3 holds. For $\rho \in (\rho^*, \min(\rho^*_1, \rho^*_2, \rho^*_3))$, a regulator examining any single channel observes $\rho < \rho^*_k$ for all $k$ and concludes that coordination, information acquisition, and liquidity provision are each well-behaved. Yet the interior fixed point is unstable: $\rho > \rho^*$ implies the spectral radius of $DT$ exceeds unity, and perturbations are amplified by the feedback loop, driving the system toward the corner fixed point characterised in Appendix F (Remark F).*

*Proof.* The single-channel assessments follow from the definitions of $\rho^*_k$. Interior instability follows from Proposition 6: $\rho > \rho^*$ implies $\lambda_+ > 1$ at the interior fixed point. The corner fixed point always exists in $K$ (Appendix F). □

Corollary 1 formalises the qualitative warning of Danielsson et al. [2022]: the interaction of the channels, not any single channel, is the source of the most dangerous systemic risk. The 1998 LTCM crisis illustrates the mechanism: the fund's distress forced liquidations that moved prices against other convergence-trade funds holding correlated positions, triggering further liquidations in a spiral that only coordinated intervention could arrest [Brunnermeier and Pedersen, 2009]. Whether the safety illusion is quantitatively large enough to matter for prudential policy depends on the width of the interval $(\rho^*, \min_i \rho_i^*)$ at empirically plausible parameters.[9]

The compound threshold result is structurally distinct from the phase-transition fragility of Acemoglu et al. [2015], though both identify parameter regions where small perturbations trigger systemic collapse. In the AOT network model, fragility arises from the topology of bilateral exposures: a monotone contagion operator propagates losses through the network, and the phase transition occurs when connectivity crosses a threshold that makes the operator's spectral radius exceed unity. Here, fragility arises from signal correlation across functionally heterogeneous channels (coordination, information acquisition, and liquidity) connected through a composite operator $T$ that is not monotone (the stabilising force $b > 0$ partially offsets the destabilising chain $h|a|$). The non-monotonicity means that the compound threshold cannot be located by the fixed-point comparison arguments available in the AOT setting; the bifurcation analysis of Proposition 6 is essential.

## 4.5 Quantitative Exercises

**Calibration.** Table 1 reports baseline values targeting US equity markets, defensible ranges, and empirical sources for all parameters entering $T$.

**Baseline results.** At calibrated parameters, the safety illusion is wide: the compound fragility threshold $\rho^* = 0.215$ lies far below $\rho_1^* = 0.671$, opening a safety illusion interval of width 0.456 (Table 2). At $\rho = 0.30$, the amplification factor reaches $2.66\times$ ($\rho_{\text{eff}} = 0.799$), placing the system well into the fragile region.

**Robustness.** The policy-relevant object is not the point estimate $\rho^* = 0.215$ but the interval $[0.11, 0.39]$ over which $\rho^*$ ranges as the $g_1$ functional form varies across concave, convex,

[9] Below $\rho^*$, the interior fixed point is locally unique by the Ostrowski theorem ($\lambda_+ < 1$ makes $T$ a local contraction); global uniqueness is not established, as corner equilibria may coexist (Appendix F). Whether $\rho^* < \min_i \rho_i^*$ extends to all fixed points of $T$ (not just the interior one) is an open question; a monotonicity argument on the fixed-point set would require showing that $T$ preserves a partial order on $K$, which is not established (see Section 4.5).

Table 1: Calibrated parameter values for the amplification loop

| Symbol | Description | Baseline | Range | Source |
|---|---|---|---|---|
| $N$ | AI-equipped dealers | 50 | [20, 200] | [a] |
| $\alpha_{\mathrm{SC}}$ | Coordination motive | 0.700 | [0.30, 1.50] | [b] |
| $p_w$ | Withdrawal probability | 0.050 | [0.01, 0.15] | [c] |
| $\sigma$ | Signal noise (std dev) | 0.150 | [0.05, 0.30] | [d] |
| $\gamma$ | Risk aversion | 2.000 | [1.0, 5.0] | [e] |
| $c_P$ | Private signal cost | 0.010 | [0.005, 0.05] | [f] |
| $k_{\mathrm{GY}}$ | GY complementarity | 0.500 | [0.1, 2.0] | † |
| $D$ | Revenue coefficient | 0.020 | [0.005, 0.05] | [g] |
| $C$ | Cost coefficient | 0.005 | [0.001, 0.02] | [g] |
| $\pi_0$ | Outside option | 0.005 | [0.001, 0.01] | [h] |
| $\tilde{m}$ | Feedback gain | 1.000 | [0.5, 2.0] | † |

*Sources:* [a] 24 FRBNY primary dealers + ∼26 HFT market makers [Menkveld, 2013]. [b] $R(\theta^*) = 1.1$ implies $\alpha_{\mathrm{SC}} = 0.69$ [Goldstein and Pauzner, 2005]. [c] He and Krishnamurthy [2013] crisis frequency; see ‡ below. [d] Quarterly equity vol (CRSP). [e] Standard macro-finance. [f] Greenwich Associates. [g] Avellaneda and Stoikov [2008]. [h] Minimum desk profit.

*Notes:* † Parameters without direct empirical anchors; flagged as sensitivity-critical in the Monte Carlo exercise. ‡ $p_w = 0.05$ calibrated to match the He and Krishnamurthy [2013] intermediary-sector capital shortfall frequency, approximately 5% annualised probability, corresponding to roughly three stress episodes per 60 years of U.S. market data, mapping to the per-period Bernoulli exit probability in Channel 3. The feedback gain $\tilde{m}$ is a scaling parameter distinct from the Jacobian derivative $m \equiv dg_3/d\mu_I$. The information acquisition elasticity $|a| \equiv |d\mu_I^*/d\theta^*|$ is not a free parameter: it is determined by the Grossman-Stiglitz indifference condition (11) and equals $c_P/[(1-\theta^*)^2 \cdot |dF/d\mu_I|]$ at the equilibrium, where $|dF/d\mu_I| = k_P/\mu_I^{*2} + k_A(1-\rho)^\alpha \beta(1-\mu_I^*)^{-\beta-1}$ (Appendix B); no additional free parameter is needed. Derived quantities at baseline: $\rho_1^* = 0.671$, $\rho^* = 0.215$, safety illusion width = 0.456.

and piecewise-linear specifications (Table IA.3 in the Internet Appendix). The linear baseline sits near the centre of this interval and serves as a benchmark interpolation. Under the convex specification the safety margin is dangerously narrow ($\rho^* = 0.105$); under the concave it is substantially wider ($\rho^* = 0.385$). Replacing $\rho$ with the Bernoulli withdrawal correlation $\rho_W(\rho, p_w)$ in $g_3$ raises $\rho^*$ from 0.215 to 0.380, narrowing the safety illusion width from 0.456 to 0.291; the safety illusion is preserved under both specifications, and the baseline $\rho$ specification is the conservative choice (Table IA.3). The baseline value $\rho^* \approx 0.21$ should therefore be interpreted as a benchmark under the linear $g_1$ and signal-level $\rho$ specifications, not a point estimate. Crucially, although the level of $\rho^*$ varies with the $g_1$ specification, the steep kink in the damage function at $\rho^*$, which drives the Weitzman instrument ranking toward quantity regulation, is invariant to $g_1$ because it arises from the saddle-node bifurcation in the GP equilibrium (Remark, Instrument Ranking Invariance, in Section 5).

A 10,000-draw Monte Carlo exercise confirms the safety illusion at all parameter combinations, with median $\rho^* = 0.055$ (10th–90th percentile: $[0.020, 0.210]$). First-order Sobol

Table 2: Amplification loop at baseline parameters

| $\rho$ (exog.) | $\rho_{\text{eff}}$ | $\theta^{**}$ | $N_{\text{eff}}$ | $\mu_I$ | Status |
|---|---|---|---|---|---|
| 0.05 | 0.198 | 0.604 | 42.2 | 0.726 | stable |
| 0.10 | 0.377 | 0.664 | 34.6 | 0.699 | stable |
| 0.15 | 0.528 | 0.722 | 27.8 | 0.670 | stable |
| 0.20 | 0.645 | 0.783 | 22.2 | 0.642 | stable |
| 0.25 | 0.734 | 0.950 | 17.8 | 0.615 | **FRAGILE** |
| 0.30 | 0.799 | 0.950 | 14.4 | 0.590 | **FRAGILE** |

*Notes:* Compound fragility threshold $\rho^* = 0.215$; Channel 1 boundary $\rho_1^* = 0.671$; safety illusion width $= 0.456$; amplification factor $2.66\times$ at $\rho = 0.30$. $N_{\text{eff}}$ uses the bivariate normal withdrawal correlation $\rho_W$ from (16). FRAGILE rows report the last computed interior fixed point; for $\rho > \rho^*$, the interior fixed point is unstable ($\lambda_+ > 1$) and the system may transition to the corner fixed point at $(1 - (1-\rho)/N,\ \theta_H,\ 1)$; see Remark F in Appendix F.

indices identify $\alpha_{\text{SC}}$ (21.4%), $N$ (16.0%), and $c_P$ (10.2%) as the key drivers, precisely those parameters with direct empirical anchors, confirming that the safety illusion does not depend on quantities lacking observational grounding. Full details, including the complete Sobol index ranking, are in the Internet Appendix (Table IA.5). The baseline $\rho^* = 0.215$ sits at the 90th percentile of the Monte Carlo distribution and is not a central estimate; it is a defensible parameterisation targeting US equity market conditions, where the coordination motive $\alpha_{\text{SC}}$ and dealer count $N$ are calibrated to specific empirical anchors rather than drawn from their full uncertainty ranges. The median $\rho^* = 0.055$ reflects the distribution over all plausible parameter combinations, many corresponding to market structures substantially different from large-cap US equities.

**Empirical mapping.** The calibrated $\rho^* \approx 0.21$ can in principle be compared to cross-sectional measures of AI signal similarity, for instance the pairwise cosine similarity of institutional portfolio changes constructed from 13F filings, or the cross-sectional correlation of I/B/E/S return forecasts. If such proxies transition from below $\rho^*$ to above it following widespread AI adoption, the model predicts that the system enters the safety illusion interval where single-channel stress tests return favourable verdicts but compound fragility is already present. The mapping is imperfect: model $\rho$ is a signal-correlation primitive, while empirical proxies also reflect common risk-factor exposures unrelated to AI. Constructing a clean supply-side measure of $\rho$, one that isolates AI-driven calibration correlation from fundamental co-movement, remains an important empirical challenge.

Formal inference on the calibrated threshold $\rho^*$ is an open methodological question. Cocci and Plagborg-Møller [2024] provide a framework for computing standard errors for calibrated parameters via influence functions; applying this framework to the multi-channel fixed-point

system, where $\rho^*$ is defined implicitly by the eigenvalue condition (25), is an important direction for future work.

In the limiting case $N \to \infty$, the compound fragility threshold $\rho^* \to 0$ (Appendix G): the amplification loop can activate at arbitrarily low levels of AI signal homogeneity. Whether competitive AI adoption can keep $\rho$ below $\rho^*$ without regulatory intervention is the subject of the next section.

**Scope of the compound threshold.** Proposition 6 and Corollary 1 are conditional on Assumption 3: the compound threshold $\rho^*$ is the loss-of-interior-stability boundary, not a global uniqueness or global fragility result. A natural extension would establish that $\rho^* < \min_i \rho_i^*$ holds for every fixed point of $T$, not just the interior one. This would require showing that $T$ preserves a componentwise partial order on $K$, that is, that $T$ is monotone in the sense of Tarski's fixed-point theorem, so that the set of fixed points is a lattice and the smallest fixed point inherits the strict inequality. The composite operator $T$ is not monotone in general: $T_1$ is decreasing in $N_{\text{eff}}$ (a higher effective market-maker count lowers effective correlation), while $T_3$ is decreasing in $\theta^*$ through the $g_2$ link (a higher crisis threshold reduces private information and lowers $N_{\text{eff}}$). These opposing monotonicity directions prevent a direct lattice-theoretic argument. Whether the compound threshold result extends beyond the interior fixed point, or whether there exist parameter configurations with multiple interior fixed points for which the safety illusion width differs, remains an open question.

# 5 Extensions

Competitive AI adoption is a prisoner's dilemma. Each fund manager's individually rational choice to adopt the dominant AI model pushes aggregate signal correlation above the compound fragility threshold, leaving all managers strictly worse off. The experimental evidence of Fish et al. [2024], where LLM-based pricing agents autonomously converge on supracompetitive strategies without explicit communication, provides direct support for this premise: shared AI architectures generate correlated behaviour even absent coordination intent. The mechanism is simple: trading rents are nearly invariant to $\rho$ (the $(1-\rho)^{0.04}$ factor from the information channel), while the career cost of deviating from the industry benchmark is first-order. This asymmetry makes adoption individually cheap and collectively catastrophic.

We model adoption as a strategic game in which each manager $i$ chooses a benchmark weight $\rho_i \in [0,1]$ on the dominant AI model. Manager $i$'s payoff has three components: expected trading profit $\pi_\alpha(\rho_i, \bar{\rho}) = k_P(1-\rho_i)^\alpha/(1+k_{\text{comp}}\bar{\rho})$, which declines weakly in own adoption and in competition; tracking error cost $TE_i = c_{\text{TE}}(1-\rho_i)\bar{\rho}$, which penalises devia-

tion from the consensus; and a systemic risk externality $SC(\bar{\rho}) = [\alpha_{\mathrm{SC}}^2/(1-\alpha_{\mathrm{SC}})^2] \cdot (\kappa_{\mathrm{sys}}/\tau) \cdot \bar{\rho}$ that depends on average adoption but not on individual choice. The systemic cost $SC(\bar{\rho})$ is the Angeletos and Pavan [2007] linear-quadratic approximation to the welfare cost of coordination risk, as derived in the welfare analysis of Section 3.1; this formula captures the qualitative direction of the externality under the linear-quadratic benchmark; in the binary-action Goldstein-Pauzner game the welfare direction can reverse at fixed crisis thresholds (Appendix D), so the welfare ordering $\rho^{\mathrm{SO}} < \rho^{\mathrm{NE}}$ should be understood as conditional on the equilibrium adjustment, which numerical verification confirms (ibid.).

The bilinear tracking error specification $TE_i = c_{\mathrm{TE}}(1-\rho_i)\bar{\rho}$ emerges from a linear benchmarking contract in delegated portfolio management. When the fund manager's compensation is $\omega_i = \bar{\omega} - \kappa \cdot |r_i - \bar{r}|$ where $r_i$ is the manager's return and $\bar{r}$ is the industry benchmark return, the expected penalty for deviating from the consensus portfolio is proportional to the distance between the manager's signal weight and the industry average, scaled by how correlated the benchmark itself is. Formally, $\mathbb{E}[|r_i - \bar{r}|] \propto (1-\rho_i)\bar{\rho}^{1/2}$ under Gaussian returns; the bilinear form $TE_i = c_{\mathrm{TE}}(1-\rho_i)\bar{\rho}$ is a first-order Taylor approximation to this penalty around $\bar{\rho} = 1$ (the linear approximation overstates the tracking error penalty by approximately 29% at $\bar{\rho} = 0.5$; using the exact formula would lower $\rho^{\mathrm{NE}}$ but preserve the qualitative prisoner's dilemma structure) [see Laffont and Martimort, 2002, for the general principal-agent framework underlying benchmarking contracts].[10]

The tracking error term drives the result. When competitors adopt ($\bar{\rho}$ high), the cost of not adopting (portfolio returns that deviate visibly from the industry) rises faster than the benefit of maintaining independent signals. This creates strategic complementarity: best responses are increasing in $\bar{\rho}$.

**Proposition 7** (Endogenous Signal Correlation)**.** (i) *The individual best response $\rho_i^*(\bar{\rho})$ is strictly increasing in $\bar{\rho}$.*

(ii) *There exists a unique stable symmetric Nash equilibrium $\rho^{NE} \in (0,1)$, characterised by*

$$\alpha k_P = c_{TE}\rho^{NE}(1-\rho^{NE})^{1-\alpha}(1 + k_{comp}\rho^{NE}). \tag{26}$$

(iii) *$\rho^{NE} > \rho^{SO}$, and under a sufficient condition on $\kappa_{sys}$, $\rho^{NE} > \rho^*$.*

(iv) *Define the cooperative outcome $\rho^{COOP}$ as the symmetric correlation level that maximises the sum of private payoffs $\sum_i[\pi_\alpha(\rho_i, \bar{\rho}) - TE_i(\rho_i, \bar{\rho})]$ subject to $\rho_i = \bar{\rho}$ for all $i$. Then $\rho^{SO} \leq \rho^{COOP} < \rho^{NE}$, and every manager strictly prefers $\rho^{COOP}$ to $\rho^{NE}$: adoption is a pris-*

[10]The strategic complementarity result is robust to a quadratic tracking error specification $TE_i = c_{\mathrm{TE}}(1-\rho_i)^2\bar{\rho}$, which arises from mean-tracking-error-variance optimisation in the standard delegated portfolio management literature. Under the quadratic form, complementarity is weaker but the Nash equilibrium, prisoner's dilemma, and welfare ordering all survive.

*oner's dilemma. The first inequality is strict whenever $SC(\rho^{COOP}) > 0$; when $\rho^{COOP} = 0$ (as occurs in the present specification, since AI adoption carries no direct cost saving beyond the tracking error motive), the two coincide and the case for mandatory regulation reduces to enforcing the cooperative solution itself.*

*Remark* (Stability of the Nash Equilibrium). The upper root of $H(\rho) = \alpha k_P$ is stable under best-response dynamics if and only if $dBR/d\bar{\rho}\big|_{\rho^{\mathrm{NE}}} < 1$. From the best-response formula in Part (i), this condition reduces to

$$\frac{(1-\rho^{\mathrm{NE}})(1+2k_{\mathrm{comp}}\rho^{\mathrm{NE}})}{(1-\alpha)\rho^{\mathrm{NE}}(1+k_{\mathrm{comp}}\rho^{\mathrm{NE}})} < 1. \tag{27}$$

For $k_{\mathrm{comp}} = 0$, this simplifies to $\rho^{\mathrm{NE}} > 1/(2-\alpha) \approx 0.51$. Since $\alpha \approx 0.04$, the tracking error motive pushes the equilibrium well above this threshold at all calibrated parameter values (see Appendix J, Part ii).

*Proof sketch.* The tracking error motive dominates the weak rent sensitivity ($\alpha \approx 0.04$), creating strategic complementarity. Because each manager ignores the systemic risk externality, $\rho^{\mathrm{NE}}$ exceeds $\rho^{\mathrm{SO}}$. The cooperative payoff satisfies $\Pi^{\mathrm{COOP}}(0) = k_P > \Pi^{\mathrm{COOP}}(\rho)$ for all $\rho > 0$ (since both the profit and TE components reduce the payoff relative to zero adoption), so $\rho^{\mathrm{COOP}} = 0$. The corner solution $\rho^{\mathrm{COOP}} = 0$ reflects the payoff specification, which assigns no direct cost saving to AI adoption; if AI reduced operational costs by $\delta$ per unit adopted, the cooperative optimum would be interior at $\rho^{\mathrm{COOP}} = \delta/k_P$, leaving the qualitative prisoner's dilemma structure intact. The social optimum coincides ($\rho^{\mathrm{SO}} = 0$) because $SC(0) = 0$. The PD structure follows: all managers prefer zero adoption to $\rho^{\mathrm{NE}}$, but each deviates because the tracking error penalty for non-adoption is first-order. Mandatory regulation enforces the cooperative solution when bilateral coordination is infeasible. Full proof in Appendix J. □

The equilibrium condition (26) balances the marginal rent loss from correlation (left-hand side, small because $\alpha \approx 0.04$) against the tracking error and competition costs of deviating from consensus (right-hand side, large). The imbalance pushes the equilibrium toward high adoption. The welfare gap in parts (iii)–(iv) immediately implies a policy remedy.

**Corollary 2** (Diversity Mandate: Local Welfare Improvement)**.** (a) *Any mandate $\rho_i \leq \rho_{\max}$ with $\rho_{\max}$ slightly below $\rho^{NE}$ strictly improves welfare. This follows directly from $dW/d\rho\big|_{\rho=\rho^{NE}} < 0$ (Proposition 7, Part iii) and requires no concavity assumption. Setting $\rho_{\max} < \rho^*$ further ensures systemic safety by placing the economy below the compound fragility threshold.*

(b) *(Conditional.) If $W$ is strictly concave on an interval containing $(\rho^{SO}, \rho^{NE})$, then $\rho^{SO}$ is the unique global maximiser and any cap $\rho_{\max} \in (\rho^{SO}, \rho^{NE})$ achieves a global welfare*

*improvement.*[11]

Why does the $SC(\bar{\rho})$ externality require mandatory regulation rather than private ordering? Two features of the setting defeat bilateral Coasian solutions. First, individual adoption levels $\rho_i$ are not directly observable by other market participants: the correlation between AI model outputs is a property of the training pipeline, not a contractible variable. Bilateral agreements to limit $\rho_i$ are therefore unverifiable. Second, $SC(\bar{\rho})$ is a public bad: all institutions bear systemic risk regardless of whether they participate in any bilateral contract, making the externality inherently non-excludable. Acemoglu et al. [2015] establish a parallel result for network-contagion externalities, where each institution's contribution to systemic risk is non-contractible because it flows through the network topology rather than through bilateral relationships. The systemic cost $SC(\bar{\rho})$ inherits this structure: it depends on the average $\bar{\rho}$, not on any bilateral pair. Leitner [2005] shows that bilateral coordination can sometimes resolve network externalities when agents can commit to mutual insurance; in the present setting, no analogous commitment device exists because the correlation $\rho_i$ is a property of the AI model, not a transferable obligation.

The prisoner's dilemma closes the loop. The same competitive forces that drive AI adoption drive the system past the compound threshold that no single-channel stress test can detect. The March 2021 collapse of Archegos Capital illustrates the mechanism in concentrated form: correlated leveraged positions across multiple prime brokers created systemic exposure that no individual counterparty priced, precisely because each broker observed only its own bilateral relationship while the aggregate concentration was visible only in the cross-section. Archegos's externality arose from leverage concentration rather than signal correlation per se, but the informational structure is identical: each counterparty's risk assessment was accurate conditional on its own exposure, yet the system-wide risk was invisible without the cross-section. A diversity mandate that caps effective signal correlation addresses this externality directly, in the same way that position limits address concentrated holdings, but targeting the information input rather than the portfolio output.

The correlation cap is not the only feasible instrument. A Pigouvian charge $SC'(\bar{\rho}) \cdot \rho_i$ on each institution's adoption level achieves the same welfare improvement under certainty; under uncertainty about the shape of $SC(\bar{\rho})$, the price instrument may dominate the quantity instrument on standard Weitzman (1974) grounds. The compound fragility threshold

[11] Global strict concavity of $W(\rho)$ is not guaranteed for all parameter values: the tracking error component contributes a convex term $2c_{\mathrm{TE}} > 0$ to $W''(\rho)$, and the competition term $1/(1 + k_{\mathrm{comp}}\rho)$ adds further convexity when $k_{\mathrm{comp}}$ is large. A sufficient condition for global concavity is $k_{\mathrm{comp}} < \alpha(1-\alpha)/(2c_{\mathrm{TE}})$, which is violated at typical calibrations (e.g., $k_{\mathrm{comp}} < 0.038$ for $c_{\mathrm{TE}} = 0.5$). The local welfare improvement claim in Part (a) is robust to this failure; the global optimality claim in Part (b) should be verified numerically for specific calibrations.

creates a steep kink in the damage function at $\rho^*$ (marginal damages are near-zero below and discontinuously large above), which tilts the Weitzman instrument ranking toward quantity regulation whenever the regulator faces uncertainty about the level of $\rho$. A structural intervention requiring AI architecture diversity, analogous to the frequent batch auction proposal of Budish et al. [2015] for high-frequency trading, would target the common-training-data source of $\rho$ directly rather than its downstream manifestation. The correlation cap is preferred if $\rho_i$ is not directly observable but aggregate adoption intensity is verifiable ex post; the Pigouvian charge is preferred if individual contributions can be assessed. The following remark establishes that the ranking between these instruments is invariant to the functional form of the $g_1$ aggregation mapping.

*Remark* (Instrument Ranking Invariance). The Weitzman instrument ranking (correlation cap dominates Pigouvian charge under regulatory uncertainty about $\rho$) is invariant to the functional form of the $g_1$ mapping from $N_{\text{eff}}$ to $\rho_{\text{eff}}$. The argument has three steps.

(i) Any monotone $g_1$ satisfying the boundary conditions $g_1(\rho, N) = \rho$ and $g_1(\rho, 1) = 1$ maps $N_{\text{eff}} \to 1$ to $\rho_{\text{eff}} \to 1 > \rho_1^*$.

(ii) As $\rho_{\text{eff}} \to \rho_1^{*-}$, the crisis sensitivity $h = d\theta^*/d\rho_{\text{eff}}$ diverges via the saddle-node bifurcation in the Goldstein-Pauzner equilibrium system (Appendix C). This divergence is a property of the GP game, not of $g_1$.

(iii) The divergence of $h$ forces the dominant eigenvalue $\lambda_+$ to cross unity at some $\rho^* < \rho_1^*$, creating a steep kink in the damage function at the bifurcation point. Under Weitzman (1974) logic, a steep kink in marginal damages (near-zero below $\rho^*$, near-infinite above) favours the quantity instrument whenever the regulator faces uncertainty about the level of $\rho$.

The level of $\rho^*$ varies across $g_1$ specifications (Table 5 reports a range of $[0.105, 0.385]$ across linear, concave, convex, and logistic forms), but all specifications produce a bifurcation and all produce a steep eigenvalue gradient at the crossing (Internet Appendix, Table IA.3). The instrument ranking is therefore robust to the acknowledged lack of microfoundation for $g_1$.

# 6 Conclusion

A single correlation primitive, the pairwise AI signal correlation $\rho$, simultaneously activates three fragility channels whose joint compound threshold lies strictly below every individual threshold. The compound threshold $\rho^* \approx 0.21$ defines a safety illusion interval in which coordination risk, information quality, and liquidity provision each appear sound while the

interior equilibrium of the integrated system has lost stability (Proposition 6). This result is conditional on the economy operating at the interior fixed point (Assumption 3); if the economy has already transitioned to a corner equilibrium with maximal fragility, single-channel assessments are moot. Competitive AI adoption is a prisoner's dilemma: career incentives drive each fund manager toward the dominant model, the Nash equilibrium signal correlation exceeds both the social optimum and $\rho^*$, and voluntary coordination cannot resolve this because signal correlation is neither observable nor contractible. A diversity mandate, whether implemented through correlation caps, model diversity requirements, or correlation-sensitive capital charges, is the minimum regulatory instrument; the operative target is the compound threshold interval $\rho^* \in [0.11, 0.39]$, reflecting sensitivity to the amplification functional form, not any individual channel threshold. Because $\rho^*$ is the interior-stability boundary, its policy relevance depends on the economy being at or near the interior equilibrium, the regime with active private information acquisition and distributed liquidity provision. This is the natural starting point for regulatory design: the interior equilibrium is locally stable for $\rho < \rho^*$, and intervention aims to prevent the transition to the fragile corner.

The multi-channel feedback that the model formalises echoes the empirical pattern documented by Brunnermeier [2009] during 2007–2008, where margin spirals, information loss, and correlated position unwinds reinforced each other. AI-driven signal homogeneity provides a more systematic source of the same cross-channel amplification. The March 2023 collapse of Silicon Valley Bank, where depositors coordinated a $42 billion withdrawal in a single day on shared beliefs about interest-rate losses [Jiang et al., 2023], illustrates that correlated beliefs can compress coordination timescales far beyond what traditional stress tests assume; AI-driven coordination, operating at algorithmic speed across shared model outputs, would be faster still.

The model's central parameter $\rho$ is not currently directly observable. The closest demand-side proxies, including portfolio-change cosine similarity from 13F filings and cross-sectional correlations of I/B/E/S return forecasts, reflect correlated trading decisions rather than the supply-side signal correlation driving the liquidity channel and the amplification loop. Three feasible supply-side proxies deserve attention: bid-ask spread co-movement across independent market makers, quote update synchronicity from TAQ order-level data, and variance-estimate correlation across data vendors [cf. Greenwood and Thesmar, 2011]. The experimental evidence of Fish et al. [2024], where LLM-based pricing agents autonomously converge on supracompetitive strategies without explicit communication, suggests that controlled experimental designs may complement observational approaches. Operationalising a clean supply-side measure of $\rho$ remains the most important empirical priority.

Three extensions define the most productive next steps. First, characterising optimal reg-

ulatory timing requires treating AI adoption as a stochastic process; this determines whether regulators have weeks or hours once the compound threshold is approached. Second, replacing the equicorrelation assumption with a full cross-model correlation matrix would connect the theory to the emerging literature on AI model clustering; systemic risk would then depend on the leading eigenvalue of this matrix rather than a scalar $\rho$. Third, integrating signal homogeneity with balance sheet network topology [Acemoglu et al., 2015] would identify when correlation-driven fragility dominates contagion-driven fragility. The information acquisition decision in the Grossman-Stiglitz channel is static; a dynamic extension in which agents learn about $\rho$ over time could generate richer amplification dynamics, particularly if agents update their beliefs about model homogeneity only after observing correlated losses.

The current trajectory of financial AI, driven by common training data, common architectures, and common outputs, is a trajectory toward higher $\rho$ [International Monetary Fund, 2024, European Securities and Markets Authority, 2025]. The evidence of Hansen and Lee [2025] that independently queried AI agents can reduce herding suggests that this trajectory is not inevitable. The question is whether regulatory frameworks will target compound fragility before the system crosses a threshold that no single-channel stress test is designed to detect.

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# A Proof of Lemma 1

The Fisher information in $(x_1, \ldots, x_N)$ about $\theta$ is $\mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, where $\Sigma$ is the equicorrelation covariance matrix with diagonal entries $\sigma^2$ and off-diagonal entries $\rho\sigma^2$. By the Sherman-Morrison formula, $\Sigma^{-1}$ has diagonal entries $(1-\rho)^{-1}\sigma^{-2}[1 - \rho N^{-1}(1 + (N-1)\rho)^{-1} \cdot N]$ and the quadratic form evaluates to

$$\mathbf{1}^\top \Sigma^{-1} \mathbf{1} = \frac{N}{\sigma^2(1-\rho)} \cdot \frac{1}{1 + (N-1)\rho/(1-\rho)} = \frac{N}{\sigma^2[1 + (N-1)\rho]}.$$

As $N \to \infty$, this converges to $1/(\rho\sigma^2) = \tau/\rho$, equal to the Fisher information of $1/\rho$ independent signals each with precision $\tau$. $\square$

# B Jacobian Derivation

At an interior fixed point $v^* = (\rho^*_{\text{eff}}, \theta^{**}, N^*_{\text{eff}})$ with $\rho^*_{\text{eff}} < \rho^*_1$, we define five partial derivatives evaluated at the fixed point. These coefficients govern the strength of each link in the feedback loop.

The first two coefficients capture how the Channel 2 information fraction responds to the other state variables:

$$a \equiv \frac{d\mu^*_I}{d\theta^*} = -\frac{c_P}{(1-\theta^*)^2 \cdot |dF/d\mu_I|} < 0, \tag{28}$$

$$b \equiv \left.\frac{d\mu^*_I}{d\rho_{\text{eff}}}\right|_{\theta^* \text{ fixed}} = \frac{\alpha k_A (1-\rho_{\text{eff}})^{\alpha-1}(1-\mu^*_I)^{-\beta}}{k_P/\mu^{*2}_I + \beta k_A (1-\rho_{\text{eff}})^{\alpha}(1-\mu^*_I)^{-\beta-1}} > 0. \tag{29}$$

Both $|a|$ and $b$ follow from the implicit function theorem applied to the Grossman-Stiglitz indifference condition (11), with

$$|dF/d\mu_I| = \frac{k_P}{\mu^{*2}_I} + \beta k_A (1-\rho_{\text{eff}})^{\alpha}(1-\mu^*_I)^{-\beta-1}. \tag{30}$$

These derivatives are fully determined by the Channel 2 primitives $(k_P, k_A, c_P, \alpha, \beta)$ evaluated at the equilibrium $(\mu^*_I, \theta^*, \rho_{\text{eff}})$; no additional free parameter is required.

The coefficient $a$ in (28) is negative because a higher crisis threshold raises the effective cost of private research, driving agents toward AI signals. The coefficient $b$ in (29) is positive because higher effective correlation reduces AI trading rents, making private information relatively more attractive.

The remaining three coefficients capture the Channel 3 liquidity link, the Channel 1 crisis sensitivity, and the $g_1$ amplification:

$$m \equiv \frac{dg_3}{d\mu_I} = \frac{2N(N-1)\rho(1-\mu_I)}{(1+(N-1)\rho(1-\mu_I)^2)^2} > 0, \tag{31}$$

$$h \equiv \left.\frac{d\theta^*}{d\rho_{\text{eff}}}\right|_{\text{IFT}} > 0 \quad \text{(in the fragile region } \rho_{\text{eff}} \in (\rho_0, \rho^*_1)\text{)}, \tag{32}$$

$$w \equiv \frac{1-\rho}{N} > 0. \tag{33}$$

The coefficient $m$ in (31) measures how the effective market-maker count responds to changes in private information acquisition: fewer privately informed agents raise market-maker reliance on the common AI signal, reducing $N_{\text{eff}}$. The coefficient $h$ in (32) is the Channel 1 crisis sensitivity: it is the implicit derivative of $\theta^*$ with respect to $\rho_{\text{eff}}$, obtained by applying the implicit function theorem to the $2 \times 2$ Goldstein-Pauzner equilibrium system $(F_1, F_2) = 0$

in $(x^*, \theta^*)$ (see Appendix C). Crucially, $h$ is *not* the scalar rate of change of the crisis threshold alone—the aggregate withdrawal condition is $\rho$-independent (Appendix D)—but rather the equilibrium response via Cramer's rule from the joint system. The coefficient $h$ diverges as $\rho_{\text{eff}}$ approaches $\rho_1^*$ because $\det(J) \to 0$ at the saddle-node bifurcation (Appendix C). Note that $h$ as defined here—the equilibrium response of $\theta^*$ to $\rho_{\text{eff}}$ via Cramer's rule from the joint $(x^*, \theta^*)$ system—coincides with the scalar derivative $d\theta^*/d\rho_{\text{eff}}$ obtained by reducing the system to one dimension via the implicit function theorem in Appendix C, equation (38). The coefficient $w$ in (33) captures the strength of the $g_1$ mapping from $N_{\text{eff}}$ to $\rho_{\text{eff}}$.

The Jacobian matrix of $T$ at $v^*$ is

$$DT = \begin{pmatrix} -wmb & -wm|a| & 0 \\ -hwmb & hwm|a| & 0 \\ mb & -m|a| & 0 \end{pmatrix}, \tag{34}$$

where the third column is zero because the updated state vector $(T_1(v), T_2(v), T_3(v))$ depends on the current $(\rho_{\text{eff}}, \theta^*)$ but not on the current $N_{\text{eff}}$ (which enters only through $g_1$, with $N'_{\text{eff}}$ computed from the current $\theta^*$ and $\rho_{\text{eff}}$). The system has rank two; stability depends on the non-trivial eigenvalues $\lambda_\pm$ from (24) of the main text. The dominant eigenvalue is $\lambda_+$ (not the trace $wm(h|a| - b)$).

# C Saddle-Node Bifurcation Verification

The Goldstein-Pauzner threshold equilibrium satisfies two conditions: an aggregate withdrawal equation $F_1 = 0$ and an indifference equation $F_2 = 0$, each in the unknowns $(x^*, \theta^*; \rho_{\text{eff}})$. Define the $2 \times 2$ Jacobian

$$J \;=\; D_{(x^*,\theta^*)}F \;=\; \begin{pmatrix} \partial F_1/\partial x^* & \partial F_1/\partial \theta^* \\ \partial F_2/\partial x^* & \partial F_2/\partial \theta^* \end{pmatrix}. \tag{35}$$

The Morris-Shin uniqueness condition is equivalent to $\det(J) \neq 0$. At $\rho_{\text{eff}} = \rho_1^*$, the uniqueness condition $\alpha_{\text{SC}}\sqrt{\rho_{\text{eff}}/(1-\rho_{\text{eff}})} = 1$ binds, so $\det(J) = 0$: the unique threshold equilibrium and an emerging second equilibrium collide in a saddle-node bifurcation. We verify the three non-degeneracy conditions of the saddle-node normal form (*cf.* Guckenheimer and Holmes, 1983, Theorem 3.4.1): (a) $\det(J) = 0$ at the bifurcation, by the Morris-Shin condition binding; (b) $\partial(\det J)/\partial\rho_{\text{eff}} \neq 0$, because $\det(J)$ is a smooth, strictly monotone function of $\alpha_{\text{SC}}^2\rho_{\text{eff}}/(1-\rho_{\text{eff}})$ near the critical value; (c) the quadratic coefficient in the reduced scalar equation is non-zero. This condition is not proved analytically; it is stated as Assumption 4

and verified numerically.

*Assumption* 4 (Saddle-Node Non-Degeneracy). At the bifurcation point $\rho_{\mathit{eff}} = \rho_1^*$, the quadratic coefficient $\partial^2 G/\partial(\theta^*)^2$ in the reduced scalar equation $G(\theta^*, \rho_{\mathit{eff}}) = 0$ is non-zero:

$$\left.\frac{\partial^2 G}{\partial(\theta^*)^2}\right|_{\rho_{\mathit{eff}}=\rho_1^*} \neq 0, \tag{36}$$

where $G(\theta^*, \rho_{\mathit{eff}}) \equiv F_2(x^*(\theta^*, \rho_{\mathit{eff}}), \theta^*; \rho_{\mathit{eff}})$ and $x^*(\theta^*, \rho_{\mathit{eff}})$ is the local solution of $F_1 = 0$ obtained by the implicit function theorem.

**Justification.** By the chain rule, the quadratic coefficient expands as

$$\frac{\partial^2 G}{\partial(\theta^*)^2} = \frac{\partial^2 F_2}{\partial(\theta^*)^2} + 2\frac{\partial^2 F_2}{\partial x^* \partial\theta^*}\frac{\partial x^*}{\partial\theta^*} + \frac{\partial^2 F_2}{\partial(x^*)^2}\left(\frac{\partial x^*}{\partial\theta^*}\right)^2 + \frac{\partial F_2}{\partial x^*}\frac{\partial^2 x^*}{\partial(\theta^*)^2}. \tag{37}$$

At the bifurcation point, $\partial F_2/\partial x^*$ involves the standard normal density $\phi$, and the second-order terms involve $\phi'(z) = -z\phi(z)$ (the Hermite function). The GP indifference condition sets $z_{\text{indiff}} = \Phi^{-1}(1 - 1/R(\theta^*))$, which equals zero only when $R(\theta^*) = 2$. For any $R(\theta^*) \neq 2$, payoff asymmetry ensures $z_{\text{indiff}} \neq 0$, so $\phi'(z_{\text{indiff}}) \neq 0$ and the curvature terms in (37) do not vanish. Economically, $R(\theta^*) = 2$ would require the bank return to exactly double the outside option—a knife-edge condition excluded by any empirically plausible calibration.

*Remark* (Numerical verification of Assumption 4). Assumption 4 has been verified numerically at all parameter configurations in the robustness range of Table 1: $\alpha_{\text{SC}} \in \{0.30, 0.50, 0.70, 1.00, 1.50\}$ and $R(\theta^*) \in \{1.02, 1.05, 1.10, 1.20, 1.50\}$ (25 configurations). At all tested calibrations, the second derivative $d^2\theta^*/d\rho_{\text{eff}}^2$ diverges in magnitude as $\rho_{\text{eff}} \to \rho_1^{*-}$, confirming the non-degeneracy. At the baseline ($\alpha_{\text{SC}} = 0.70$, $R = 1.1$), the estimated saddle-node constant is $C \approx 1.02$ with coefficient of variation 0.12 across distances from the bifurcation. The only degenerate case is $R(\theta^*) = 2$ (symmetric payoffs), which lies outside the empirically relevant range.

By the implicit function theorem applied to $F_1$, we can locally express $x^* = x^*(\theta^*, \rho_{\text{eff}})$ for $\rho_{\text{eff}} < \rho_1^*$, reducing the system to the scalar equation $G(\theta^*, \rho_{\text{eff}}) = 0$. Under Assumption 4, the saddle-node conditions (a)–(c) imply the standard local expansion (Guckenheimer and Holmes, 1983, Theorem 3.4.1):

$$\theta^*(\rho_{\text{eff}}) = \theta^*(\rho_1^*) - C\sqrt{\rho_1^* - \rho_{\text{eff}}} + O(\rho_1^* - \rho_{\text{eff}}), \quad C > 0, \tag{38}$$

so that $h = d\theta^*/d\rho_{\text{eff}} = C/(2\sqrt{\rho_1^* - \rho_{\text{eff}}}) + O(1) \to \infty$ as $\rho_{\text{eff}} \to \rho_1^{*-}$. The local expansion (38) establishes that $h$ diverges at the square-root rate characteristic of saddle-node bifurcations,

driving the amplification result in Proposition 6.

# D Proofs for Coordination Failure Propositions

## D.1 Proof of Proposition 2

*Aggregate condition is $\rho$-independent.* By the probit convolution identity, integrating the AI withdrawal fraction over the common shock $\eta \sim \mathcal{N}(0,1)$ yields

$$\mathbb{E}_\eta[\ell_{\mathrm{AI}}(\theta, \eta, x^*)] = \Phi\left(\frac{x^* - \theta}{\sigma}\right), \tag{39}$$

which is independent of $\rho$. The common shock component $\sqrt{\rho}\,\eta$ integrates out exactly. The total signal variance is $\sigma^2 = \rho\sigma^2 + (1-\rho)\sigma^2$ regardless of the decomposition, so the aggregate withdrawal condition does not vary with $\rho$. This rules out the "competing effects" explanation.

*Non-monotonicity through higher-order beliefs.* The non-monotonicity operates through the individual depositor's indifference condition. At the indifference threshold, an agent with signal $x_i = x^*$ must be indifferent between withdrawing and staying:

$$R(\theta^*) \cdot \Pr[\text{bank survives} \mid x_i = x^*] = 1. \tag{40}$$

Conditional on $x_i = x^*$ and $\theta = \theta^*$, the common shock satisfies $\eta \mid x_i, \theta \sim \mathcal{N}(\sqrt{\rho}(x_i - \theta)/\sigma^2,\ (1-\rho)\sigma^2)$. As $\rho$ increases, $\mathrm{Var}(\eta \mid x_i, \theta) = (1-\rho)\sigma^2$ falls: agents become more certain about the common shock. At low $\rho$, idiosyncratic uncertainty creates an option value of staying (stabilising). At high $\rho$, reduced uncertainty means a bad signal reliably predicts coordinated withdrawal; with $R(\theta) > 1$, this is self-fulfilling, raising $\theta^*$. With $R = 1$, the indifference condition is $\rho$-independent, confirming that $R(\theta) > 1$ is necessary. By the implicit function theorem applied to (40), $d\theta^*/d\rho$ exists and is continuous for $\rho < \rho_1^*$, with sign reversal at $\rho_0$. □

*Numerical verification.* The non-monotonicity is quantitatively small (approximately 0.001 in $\theta^*$) but structurally robust: for $\sigma = 0.05$, $R = 1.5$, $\lambda = 1$, $\theta^*(\rho)$ attains a minimum near $\rho \approx 0.04$ and the U-shape persists across a wide range of parameter values.

## D.2 Welfare Analysis

The Angeletos and Pavan [2007] (Proposition 4) welfare result states that in a coordination game with strategic complementarity $\alpha$, agents place weight $\alpha/(1-\alpha)$ on common informa-

tion relative to the efficient benchmark, creating a welfare loss proportional to the square of this over-weighting. The variance of the common action component is $\lambda^2 \cdot \rho\sigma^2$, giving welfare loss (9). The private benefit of correlation is the reduction in idiosyncratic decision error: as $\rho$ increases, the private noise $(1-\rho)\sigma^2$ falls. The social cost exceeds the private benefit when $[\alpha_{\text{SC}}^2/(1-\alpha_{\text{SC}})^2] \cdot \lambda^2 > 1$, which holds for $\alpha_{\text{SC}}$ and $\lambda$ above threshold values.

*Approximation caveat.* The Angeletos and Pavan [2007] framework assumes continuous actions and quadratic payoffs; the Goldstein-Pauzner game has binary actions and piecewise payoffs. At a fixed crisis threshold, welfare in the binary game can increase with $\rho$, opposite to the A-P prediction. The total welfare effect through equilibrium adjustment numerically confirms the qualitative direction, but the A-P formula should be read as a linear-quadratic benchmark. □

# E Proofs for Information Channel Results

## E.1 Proof of Lemma 2

*Part (i):* From the indifference condition $\pi_P(\mu_I^*) - c_P = \pi_A(\rho, 1-\mu_I^*)$. Since $d\pi_A/d\rho < 0$ (via the price informativeness channel), the right-hand side falls in $\rho$, albeit weakly ($\alpha \approx 0.04$). For equality to hold, $\pi_P$ must also fall, requiring $\mu_I^*$ to rise. This within-channel substitution effect is quantitatively modest given the weak $\rho$-dependence of AI rents.

*Part (ii):* Incorporating crisis risk, the indifference condition becomes $\pi_P - \pi_A = c_P/(1-\theta^*)$. Since $d\theta^*/d\rho > 0$ in the fragile region, the right-hand side rises in $\rho$, requiring a larger gap, achievable by lowering $\mu_I^*$; this cross-channel effect dominates in the fragile region.

*Part (iii):* Direct computation using the information collapse formula: $N_{\text{info}}(\rho, \mu_I) = \mu_I N + (1-\mu_I)N/(1+((1-\mu_I)N-1)\rho)$. The second term is strictly decreasing in $\rho$ for any fixed $\mu_I > 0$ and $N > 1$. □

## E.2 Proof of Proposition 3

*Part (i):* $\text{FPE} = \tau_{\text{price}}/(\tau_v + \tau_{\text{price}})$. At low $\rho$, cheap AI signals contribute diverse information, raising $\tau_{\text{price}}$. At high $\rho$, information collapse reduces the AI contribution faster than private-information substitution compensates, eventually lowering FPE.

*Part (ii):* RPE is the novel information content of the price beyond the common AI signal. The total derivative of RPE in $\rho$ has a direct effect (the second term in (13) is strictly decreasing) and an indirect effect through $\mu_I^*$. The indirect effect has sign depending on whether $\mu_I^*$ is above or below $(1-\rho)\tau/(2\tau_P)$. Under the stated sufficient condition, the direct effect dominates and RPE is monotonically decreasing. This holds for $c_P > 0$ at

typical parameter values; the near-constancy of $\mu_I^*$ in $\rho$ (from the weak $\rho$-dependence of AI rents) further simplifies verification.

*Part (iii):* FPE measures total prediction accuracy including information already available via the common AI signal; RPE measures only the novel component. The divergence follows from the definitions. □

# F Local Uniqueness Below the Threshold

**Proposition 8** (Local Uniqueness Below the Threshold)**.** *For $\rho < \rho^*$, the dominant eigenvalue satisfies $\lambda_+ < 1$, so the spectral radius $\varrho(DT) = \lambda_+ < 1$. By the Ostrowski theorem (see, e.g., Ortega and Rheinboldt, 1970, Theorem 10.1.3), there exists a matrix norm $\|\cdot\|$ and a neighbourhood $U \subset K$ of the fixed point such that $\|DT(v)\| < 1$ for all $v \in U$, making $T$ a local contraction. By the Banach fixed-point theorem, the interior fixed point is locally unique within $U$.*

*Remark* (Global uniqueness and the corner fixed point)*.* Global uniqueness of the interior fixed point is not established. The Brouwer theorem guarantees existence of at least one fixed point in $K$ but not uniqueness; and the Ostrowski local contraction does not extend globally. For $\rho > \rho^*$, the interior fixed point loses stability ($\lambda_+ > 1$) and the system may transition to a *corner fixed point.*

**Corner fixed point characterisation.** The corner fixed point $v^c = (\rho^c_{\text{eff}}, \theta^c, N^c_{\text{eff}})$ corresponds to the maximal-fragility absorbing state:

(i) $N^c_{\text{eff}} = g_3(\mu_I = 0, \rho) = N/(1 + (N-1)\rho)$ (the effective market-maker count when all agents rely on the common AI signal; this equals 1 only at $\rho = 1$ and exceeds 1 for any $\rho < 1$);

(ii) $\rho^c_{\text{eff}} = 1 - (1-\rho)N^c_{\text{eff}}/N = 1 - (1-\rho)/(1 + (N-1)\rho)$ (maximal effective correlation given exogenous $\rho$ and $N^c_{\text{eff}}$ from (i), via equation (20));

(iii) $\theta^c = \theta_H$ (worst-case crisis threshold, at the upper boundary of the GP equilibrium range).

At $v^c$, the economy operates with extreme fragility: near-minimal effective liquidity provision, near-maximal effective correlation, and the highest possible crisis threshold. This state is self-reinforcing: with $\mu_I = 0$, no agent finds it profitable to acquire private information (the crisis threshold is so high that the cost exceeds any informational advantage),

and with $N_{\text{eff}}^c = N/(1+(N-1)\rho)$ close to 1 for moderate $\rho$, the liquidity concentration is near-maximal. The corner fixed point always exists in $K$ and is a fixed point of $T$ by the boundary conditions of $g_1$, $g_2$, and $g_3$.

**Interpretation.** The interior fixed point represents normal market functioning with positive private information acquisition and distributed liquidity provision. The corner fixed point represents the post-fragility state. The compound fragility threshold $\rho^*$ marks the boundary between these regimes: for $\rho < \rho^*$, the interior fixed point is locally stable; for $\rho > \rho^*$, perturbations from the interior fixed point are amplified by the feedback loop ($\lambda_+ > 1$), driving the system toward the corner. The analogy to the robust-yet-fragile property of interconnected financial networks [Acemoglu et al., 2015] is direct: below $\rho^*$ the system absorbs shocks efficiently; above it, the same interconnection structure propagates shocks catastrophically.

# G Large Market Limit

The bifurcation analysis in the preceding sections characterises $\rho^*$ for finite $N$. For large $N$, $\rho^*(N) \to 0$ via an existence mechanism rather than the finite-$N$ Jacobian analysis. At any interior fixed point with $\mu_I < 1$, the effective correlation satisfies

$$\rho_{\text{eff}} = 1 - (1-\rho)\frac{N_{\text{eff}}}{N} = 1 - \frac{(1-\rho)}{1+(N-1)\rho(1-\mu_I)^2} \to 1 \quad \text{as } N \to \infty. \tag{41}$$

Since $\rho_{\text{eff}} \to 1 > \rho_1^*$ for any $\rho_1^* < 1$, no interior fixed point with $\rho_{\text{eff}} < \rho_1^*$ exists for sufficiently large $N$. The interior fixed point exists only for $N < N_0(\rho)$, where

$$N_0 = \frac{\rho_1^* - \rho}{(1-\rho_1^*)\cdot\rho\cdot(1-\mu_I^*)^2} + 1. \tag{42}$$

For $N > N_0(\rho)$, no stable interior fixed point exists, implying $\rho^*(N) \to 0$ as $N \to \infty$. Numerical verification: with $\alpha_{\text{SC}} = 0.8$ ($\rho_1^* = 0.61$) and $\mu_I = 0.3$, $\rho^*(10) = 0.22$, $\rho^*(100) = 0.031$, $\rho^*(1000) = 0.003$, approaching zero monotonically.

# H Proof of Proposition 6

## H.1 Full Proof

*Step 1 (Individual channel stability).* For $\rho < \rho_i^*$ considered individually, each channel maintains a stable interior equilibrium. With cross-channel feedback severed, the Jacobian is zero and the spectral radius is zero.

*Step 2 (Divergence of $h$ and positive spectral radius).* In the coupled system, the dominant eigenvalue is $\lambda_+$ from (24). The Goldstein-Pauzner equilibrium system undergoes a saddle-node bifurcation at $\rho_{\text{eff}} = \rho_1^*$, where the $2 \times 2$ Jacobian of the equilibrium conditions satisfies $\det(J) = 0$ (see (35)). Under Assumption 4, the three non-degeneracy conditions of the saddle-node normal form (Guckenheimer and Holmes, 1983, Theorem 3.4.1) hold, and the local expansion (38) yields $h = C/(2\sqrt{\rho_1^* - \rho_{\text{eff}}}) + O(1) \to \infty$ as $\rho_{\text{eff}} \to \rho_1^{*-}$. As $h \to \infty$, the discriminant in (24) grows as $8h|a|b$, so $\lambda_+ \sim wm\sqrt{h|a|b} \to \infty$. The crossing $\lambda_+ = 1$ occurs for finite $h$ (at $\rho_{\text{eff}} < \rho_1^*$), confirming the bifurcation exists. Since $\lambda_+ > wm(h|a| - b)$ whenever $b > 0$, the bifurcation occurs at a *lower* $\rho$ than the trace-based formula predicts, strengthening the result.

*Step 3 (Strict inequality $\rho^* < \min_i \rho_i^*$).* Amplification from $g_1$ implies $\rho_{\text{eff}}^* > \rho$ whenever $N_{\text{eff}}^* < N$. Define $\rho^*$ as the solution to $\rho_{\text{eff}}^*(\rho^*) = \rho_1^*$; since $\rho_{\text{eff}}^* > \rho$, we obtain $\rho^* < \rho_1^*$. For Channels 2 and 3: the coupled system's effective correlation $\rho_{\text{eff}}^* > \rho$ and elevated crisis threshold $\theta^{**} > \theta^*(\rho)$ both tighten the indifference and participation constraints beyond what exogenous $\rho$ alone produces, ensuring $\rho^* < \rho_2^*$ and $\rho^* < \rho_3^*$. □

# I Derivation of Complementarity Formula

Motivated by the Goldstein and Yang [2015] multi-dimensional information complementarity mechanism, when AI homogenises one information dimension (agents receive signals about dimension $d_1$ with correlation $\rho$), the complementarity in information acquisition about dimension $d_2$ is approximated by:

$$C_{\text{GY}}(\rho) = C_{\text{GY}}(0) \cdot \frac{1-\rho}{1-\rho+\rho k_{\text{GY}}}, \tag{43}$$

where $k_{\text{GY}} = \gamma^2 \tau_{d_1}/(\tau_{d_2}\sigma_u^2) > 0$ captures the cross-dimensional information sensitivity in the GY framework (with $\tau_{d_1}$, $\tau_{d_2}$ the signal precisions for each dimension). The formula (43) is a reduced-form characterisation motivated by the GY mechanism, not derived from the full equilibrium. Direct substitution confirms three properties: at $\rho = 0$, the formula recovers $C_{\text{GY}}(0)$, the full complementarity level at independence; at $\rho = 1$, $C_{\text{GY}}(1) = 0$ (no comple-

mentarity under perfect correlation); and $dC_{\mathrm{GY}}/d\rho = -C_{\mathrm{GY}}(0) \cdot k_{\mathrm{GY}}/(1-\rho+\rho k_{\mathrm{GY}})^2 < 0$ (strictly decreasing). The qualitative properties—strictly decreasing in $\rho$ with these boundary values—are robust to the functional form.

# J Proof of Proposition 7

*Part (i):* The best response solves $\max_{\rho_i} \pi_\alpha(\rho_i, \bar{\rho}) - TE_i(\rho_i, \bar{\rho})$. The first-order condition with exponent $\alpha$ yields

$$-\frac{\alpha k_P (1-\rho_i)^{\alpha-1}}{1+k_{\mathrm{comp}}\bar{\rho}} + c_{\mathrm{TE}}\bar{\rho} = 0,$$

so the best response is $\rho_i^*(\bar{\rho}) = 1-\Gamma(\bar{\rho})^{1/(1-\alpha)}$, where $\Gamma(\bar{\rho}) = \alpha k_P/(c_{\mathrm{TE}}\bar{\rho}(1+k_{\mathrm{comp}}\bar{\rho}))$ is strictly decreasing in $\bar{\rho}$. Since $\alpha < 1$, the exponent $1/(1-\alpha) > 1$, and $\rho_i^*$ is strictly increasing in $\bar{\rho}$: AI adoption is a strategic complement.

*Part (ii):* The Nash equilibrium solves $\rho = \rho_i^*(\rho)$, giving (26). The function $H(\rho) = c_{\mathrm{TE}}\rho(1-\rho)^{1-\alpha}(1+k_{\mathrm{comp}}\rho)$ satisfies $H(0) = H(1) = 0$ with a unique interior maximum. For $\alpha k_P$ below the maximum, $H(\rho) = \alpha k_P$ has two roots; the upper root is stable under best-response dynamics. The stability condition requires $dBR/d\bar{\rho}|_{\rho^{\mathrm{NE}}} < 1$. From the best-response formula in Part (i), at the Nash equilibrium where $\Gamma^{1/(1-\alpha)} = 1-\rho^{\mathrm{NE}}$:

$$\left.\frac{dBR}{d\bar{\rho}}\right|_{\rho^{\mathrm{NE}}} = \frac{(1-\rho^{\mathrm{NE}})(1+2k_{\mathrm{comp}}\rho^{\mathrm{NE}})}{(1-\alpha)\rho^{\mathrm{NE}}(1+k_{\mathrm{comp}}\rho^{\mathrm{NE}})} < 1. \tag{44}$$

For $k_{\mathrm{comp}} = 0$, this reduces to $\rho^{\mathrm{NE}} > 1/(2-\alpha) \approx 0.51$. Given $\alpha \approx 0.04$ and the tracking error motive pushing the equilibrium toward high adoption, this condition is satisfied at calibrated parameter values. Because $\alpha \approx 0.04$, $H$ is nearly flat near 0, pushing the Nash equilibrium well above 0.51.

*Part (iii):* The social planner maximises $W(\rho) = \pi_\alpha(\rho,\rho) - TE(\rho,\rho) - SC(\rho)$, internalising the systemic risk externality and the competition externality. At $\rho^{\mathrm{NE}}$, three additional negative terms (competition, tracking error, and systemic risk externalities) make social marginal cost exceed social marginal benefit, so $\rho^{\mathrm{SO}} < \rho^{\mathrm{NE}}$.

*Part (iv):* Define the cooperative payoff as the sum of private payoffs evaluated at a common adoption level: $\Pi^{\mathrm{COOP}}(\rho) = \pi_\alpha(\rho,\rho) - TE(\rho,\rho) = k_P(1-\rho)^\alpha/(1+k_{\mathrm{comp}}\rho) - c_{\mathrm{TE}}(1-\rho)\rho$. The cooperative outcome $\rho^{\mathrm{COOP}}$ maximises $\Pi^{\mathrm{COOP}}(\rho)$ on $[0,1]$.

$\rho^{COOP} = 0 < \rho^{NE}$*:* We show $\Pi^{\mathrm{COOP}}(0) > \Pi^{\mathrm{COOP}}(\rho)$ for all $\rho \in (0,1]$. At $\rho = 0$: $\Pi^{\mathrm{COOP}}(0) = k_P$. At $\rho = 1$: $\Pi^{\mathrm{COOP}}(1) = 0$ (since $(1-1)^\alpha = 0$ for $\alpha > 0$). For $\rho \in (0,1)$, $\Pi^{\mathrm{COOP}}(\rho) = k_P(1-\rho)^\alpha/(1+k_{\mathrm{comp}}\rho) - c_{\mathrm{TE}}(1-\rho)\rho < k_P$ since both the profit term satisfies $(1-\rho)^\alpha/(1+k_{\mathrm{comp}}\rho) < 1$ and the TE term is non-negative. Hence $\rho^{\mathrm{COOP}} = 0$:

the cooperative solution is zero AI adoption.[12] The individual Nash first-order condition, by contrast, balances the marginal rent loss $\alpha k_P(1-\rho)^{\alpha-1}/(1+k_{\text{comp}}\bar{\rho})$ against the tracking error pressure $c_{\text{TE}}\bar{\rho}$, yielding $\rho^{\text{NE}} \gg 0$ because each manager takes $\bar{\rho}$ as given. The cooperative-Nash gap arises because the individual ignores two externalities: (a) the competition effect through $k_{\text{comp}}\bar{\rho}$, and (b) the tracking error externality, where each manager's adoption raises $\bar{\rho}$ and thereby imposes career risk on non-adopters.

*$\rho^{SO} \leq \rho^{COOP}$:* The social welfare function is $W(\rho) = \Pi^{\text{COOP}}(\rho) - SC(\rho)$, where $SC(\rho) \geq 0$ for all $\rho \geq 0$ and $SC(0) = 0$. Since $\Pi^{\text{COOP}}$ achieves its maximum at $\rho^{\text{COOP}} = 0$ and $SC(\rho) \geq 0$ everywhere, we have $W(\rho) \leq \Pi^{\text{COOP}}(\rho) \leq \Pi^{\text{COOP}}(0) = W(0)$, so $\rho^{\text{SO}} = 0 = \rho^{\text{COOP}}$. The inequality is weak because $SC(0) = 0$: at the cooperative optimum, the systemic cost is already zero, so the social planner and the cooperative coalition agree. For specifications in which AI adoption carries a direct cost saving (making $\Pi^{\text{COOP}}$ non-monotone with an interior maximum $\rho^{\text{COOP}} > 0$), the $SC$ externality would drive $\rho^{\text{SO}}$ strictly below $\rho^{\text{COOP}}$, creating a gap that private coordination cannot close.

*Prisoner's dilemma structure:* Every manager strictly prefers $\rho^{\text{COOP}}$ to $\rho^{\text{NE}}$. Since $\Pi^{\text{COOP}}(0) = k_P > \Pi^{\text{COOP}}(\rho^{\text{NE}})$ (as established above), we have $\Pi^{\text{COOP}}(\rho^{\text{COOP}}) > \Pi^{\text{COOP}}(\rho^{\text{NE}})$. At any symmetric outcome $\rho_i = \bar{\rho}$ for all $i$, each manager's individual payoff equals $\Pi^{\text{COOP}}(\bar{\rho})/N$ (by symmetry), so each manager is strictly better off at $\rho^{\text{COOP}}$ than at $\rho^{\text{NE}}$. Yet no individual manager has an incentive to deviate to $\rho^{\text{COOP}}$ unilaterally: at $\bar{\rho} = \rho^{\text{NE}}$, the best response is $\rho^{\text{NE}}$ by definition of the Nash equilibrium. This is the prisoner's dilemma: the tracking error externality traps all managers at the Nash outcome even though unanimous zero-adoption would make each strictly better off.

*Mandatory regulation:* In the present specification, private coordination suffices to reach the social optimum (since $\rho^{\text{SO}} = \rho^{\text{COOP}} = 0$), but coordination is blocked by the prisoner's dilemma. Mandatory regulation (a binding cap on $\rho_i$) is the instrument that enforces the cooperative solution when bilateral coordination is infeasible (see Section 5 for the non-contractibility and public-bad arguments). □

# K Proofs for Liquidity Channel Results

## K.1 Proof of Lemma 3

The result rests on the monotonicity and convexity of the withdrawal correlation $\rho_W(\rho, p_w)$ in $\rho$.

[12] The function $\Pi^{\text{COOP}}$ is not globally monotone: the tracking error component $c_{\text{TE}}(1-\rho)\rho$ has derivative $c_{\text{TE}}(1-2\rho)$, which is negative for $\rho > 1/2$, allowing $\Pi^{\text{COOP}}$ to have a local minimum near $\rho \approx 0.5$. But the global maximum remains at $\rho = 0$ since $\Pi^{\text{COOP}}(0) = k_P > \Pi^{\text{COOP}}(\rho)$ for all $\rho > 0$.

*Monotonicity of $\rho_W$.* Differentiating the bivariate normal CDF $\Phi_2(-c, -c; \rho)$ with respect to $\rho$ yields the standard result $\partial\Phi_2(a, b; \rho)/\partial\rho = \phi_2(a, b; \rho) > 0$, where $\phi_2$ is the bivariate normal density. Since $p_w$ and hence $c = \Phi^{-1}(1 - p_w)$ are independent of $\rho$, the numerator of $\rho_W$ in (16) is strictly increasing in $\rho$, and the denominator $p_w(1 - p_w)$ is constant, so $d\rho_W/d\rho > 0$.

*Boundary conditions.* At $\rho = 0$, the bivariate normal CDF factors: $\Phi_2(-c, -c; 0) = \Phi(-c)^2 = p_w^2$, giving $\rho_W(0, p_w) = 0$ and $N_{\text{eff}}(0, p_w) = N$. At $\rho = 1$, withdrawal events are identical across agents: $\Phi_2(-c, -c; 1) = \Phi(-c) = p_w$, giving $\rho_W(1, p_w) = (p_w - p_w^2)/(p_w(1 - p_w)) = 1$ and $N_{\text{eff}}(1, p_w) = 1$.

*Parts (i)–(ii).* Part (i) follows from the boundary conditions. For part (ii), since $dN_{\text{eff}}/d\rho = -N(N-1)(d\rho_W/d\rho)/(1 + (N-1)\rho_W)^2$ and $d\rho_W/d\rho > 0$, we have $dN_{\text{eff}}/d\rho < 0$.

*Part (iii): Convexity.* Write $N_{\text{eff}} = N/(1 + (N-1)\rho_W)$. By the chain rule,

$$\frac{d^2 N_{\text{eff}}}{d\rho^2} = \frac{N(N-1)}{(1 + (N-1)\rho_W)^3} \left[ 2(N-1)\left(\frac{d\rho_W}{d\rho}\right)^2 - (1 + (N-1)\rho_W)\frac{d^2\rho_W}{d\rho^2} \right].$$

The second derivative of $\Phi_2(-c, -c; \rho)$ with respect to $\rho$ involves the derivative of the bivariate normal density, which for symmetric arguments $a = b = -c$ satisfies $d\phi_2/d\rho \leq 0$ for $\rho \in [0, 1)$ when $c > 0$ (i.e., $p_w < 0.5$). Hence $d^2\rho_W/d\rho^2 \leq 0$: the withdrawal correlation is concave in $\rho$. Since $d\rho_W/d\rho > 0$ and the first term in brackets is strictly positive while the second is non-positive, the expression in brackets is positive, giving $d^2 N_{\text{eff}}/d\rho^2 > 0$. □

## K.2 Regularity Conditions for Proposition 4

The no-equilibrium threshold $\rho^{**}$ in (18) is well-defined and lies in $(0, 1)$ under three conditions on the revenue coefficient $D = Qs_0/N^2$, the quadratic cost coefficient $C = \gamma\sigma_V^2\kappa_{\text{inv}}$, and the outside option $\pi_0 > 0$:

(R1) $D^2 \geq 4C\pi_0$. The participation constraint $Dz - Cz^2 = \pi_0$ has real roots. Economically, the maximum attainable net profit $D^2/(4C)$ must cover the outside option.

(R2) $D - C > \pi_0$. At $z = 1$ (the independent-market-maker case, $\rho_W = 0$), net profit exceeds the outside option. This ensures the equilibrium exists at low correlation.

(R3) $D + \sqrt{D^2 - 4C\pi_0} < 2CN$. The upper root $z^{**}$ satisfies $z^{**} < N$, equivalently $\rho^{**} < 1$. This ensures the threshold lies within the admissible range rather than being vacuous.

When all three hold, the quadratic $Dz - Cz^2 - \pi_0$ is positive for $z \in (z_-, z^{**})$ and negative for $z > z^{**}$, confirming that $\rho^{**}$ is a genuine upper bound on the existence of a profitable market-making equilibrium.

## K.3 Derivation of the No-Equilibrium Threshold

Revenue per market maker scales linearly with the variance inflation factor $z = 1+(N-1)\rho_W$: each unit increase in effective correlation widens the equilibrium spread proportionally, giving $\mathrm{Rev}(\rho) = D \cdot z$. Inventory cost, by contrast, scales with the *variance* of the aggregate correlated position. A portfolio of $N$ market-maker positions with pairwise withdrawal correlation $\rho_W$ has variance proportional to $z^2$ rather than $z$, reflecting the standard result that margin requirements and risk charges scale with position variance [Brunnermeier and Pedersen, 2009]. The cost specification is therefore $\mathrm{Cost}(\rho) = C \cdot z^2$.

Net profit per market maker is

$$\Pi(z) = D \cdot z - C \cdot z^2 - \pi_0, \tag{45}$$

a downward-opening quadratic in $z$ (since $C > 0$). The profit is hump-shaped: it peaks at $z_{\text{peak}} = D/(2C)$, corresponding to $\rho_{\text{peak}} = (D/(2C)-1)/(N-1)$, and declines on either side. Setting $\Pi(z) = 0$ and solving:

$$z = \frac{D \pm \sqrt{D^2 - 4C\pi_0}}{2C}.$$

Under condition (R1), both roots are real. The lower root $z_-$ is the entry threshold; the upper root $z^{**}$ is the exit threshold. Converting to $\rho$ via $\rho^{**} = (z^{**}-1)/(N-1)$ yields (18). For $\rho > \rho^{**}$, the inventory cost of holding correlated positions dominates revenue from wider spreads, and no market maker can profitably remain. □

Table 3: One-parameter comparative statics for $\rho^*$

| Parameter | Value | $\rho^*$ | $\rho_1^*$ | Illusion Width | Amp. Factor |
|---|---|---|---|---|---|
| $N$ | 20 | 0.370 | 0.671 | 0.301 | 1.93 |
| | 50 [**base**] | 0.215 | 0.671 | 0.456 | 2.66 |
| | 100 | 0.130 | 0.671 | 0.541 | 3.03 |
| | 200 | 0.070 | 0.671 | 0.601 | 3.21 |
| $\alpha_{\text{SC}}$ | 0.300 | 0.475 | 0.917 | 0.442 | 2.66 |
| | 0.700 [**base**] | 0.215 | 0.671 | 0.456 | 2.66 |
| | 1.000 | 0.140 | 0.500 | 0.360 | 2.66 |
| | 1.500 | 0.080 | 0.308 | 0.228 | 2.66 |
| $c_P$ | 0.005 | 0.380 | 0.671 | 0.291 | 1.70 |
| | 0.010 [**base**] | 0.215 | 0.671 | 0.456 | 2.66 |
| | 0.020 | 0.145 | 0.671 | 0.526 | 2.91 |
| | 0.050 | 0.100 | 0.671 | 0.571 | 3.03 |

*Notes:* Each row varies one parameter, holding others at baseline. $\rho^*$ is decreasing in all three parameters. Amplification factor $= \rho_{\text{eff}}/\rho$ at $\rho = 0.30$. The withdrawal probability $p_w$ does not affect $\rho^*$ in the baseline specification because $g_3$ uses signal-level correlation $\rho$ rather than the withdrawal correlation $\rho_W$; under the $\rho_W$ specification, $p_w$ would enter through the Bernoulli attenuation and affect the compound fragility threshold. The amplification factor at $\rho = 0.30$ is invariant to $\alpha_{\text{SC}}$ because it depends on $N_{\text{eff}}$ through $g_3$, not on $\alpha_{\text{SC}}$, which enters only through the threshold location $\rho^*$.

# Internet Appendix

This appendix contains supplementary calibration exercises supporting the results in Section 4.5. All tables and figures below use the parameter values from Table 1 unless otherwise noted.

## IA.1 Comparative Statics

Table 3 reports one-parameter comparative statics for the three most important parameters identified by the Sobol analysis: $N$, $\alpha_{\text{SC}}$, and $c_P$.

Three monotonicity results emerge. First, $\rho^*$ is strictly decreasing in $N$: more market makers strengthen the amplification loop through $g_3$, lowering the compound fragility threshold from 0.370 ($N = 20$) to 0.070 ($N = 200$). Second, $\rho^*$ is strictly decreasing in $\alpha_{\text{SC}}$: stronger strategic complementarity lowers both $\rho^*$ and $\rho_1^*$, but the illusion width is non-monotone. Third, $\rho^*$ is strictly decreasing in $c_P$: costlier private information suppresses the stabilising Channel 2 offset.

Table 4: Interaction table: $\rho^*$ as a function of $N$ and $\alpha_{\text{SC}}$

| $N \setminus \alpha_{\text{SC}}$ | 0.300 | 0.500 | 0.700 | 1.000 |
|---|---|---|---|---|
| 20 | 0.675 | 0.490 | 0.370 | 0.255 |
| 50 | 0.475 | 0.305 | 0.215 | 0.140 |
| 100 | 0.315 | 0.185 | 0.130 | 0.085 |
| 200 | 0.190 | 0.105 | 0.070 | 0.045 |

*Notes:* $\rho^*$ is decreasing in both $N$ and $\alpha_{\text{SC}}$. The interaction term is $-0.050$ (slightly sub-additive), indicating that the marginal effect of each parameter weakens as the other increases.

Table 5: Robustness of $\rho^*$ across alternative specifications

| Specification | $\rho^*$ | Illusion Width | Preserved? |
|---|---|---|---|
| $g_1$ linear (baseline) | 0.215 | 0.456 | Y |
| $g_1$ concave ($\sqrt{\cdot}$) | 0.385 | 0.286 | Y |
| $g_1$ convex ($(\cdot)^2$) | 0.105 | 0.566 | Y |
| $g_1$ logistic | 0.180 | 0.491 | Y |
| Cost: quadratic (baseline) | 0.215 | 0.456 | Y |
| Cost: power-1.5 | 0.215 | 0.456 | Y |
| $g_3$ uses $\rho_W$ (alt. Channel 3 spec.) | 0.380 | 0.291 | Y |

*Notes:* "Preserved?" indicates whether $\rho^* < \rho_1^*$ (positive safety illusion width). All seven specifications preserve the safety illusion. $\rho^*$ ranges from 0.105 to 0.385 across the $g_1$ and cost alternatives. The final row replaces the signal-level $\rho$ in $g_3$ with the Bernoulli withdrawal correlation $\rho_W(\rho, p_w)$ from (16): because $\rho_W < \rho$ for $p_w = 0.05$, effective correlation is attenuated, $N_{\text{eff}}$ rises, and the bifurcation threshold increases from 0.215 to 0.380. Under this specification, $p_w$ enters the amplification loop through $\rho_W$ and its first-order Sobol index for $\rho^*$ rises from near zero to approximately 8%, while the safety illusion is preserved. The baseline specification (using $\rho$ in $g_3$) is the conservative choice.

## IA.2 Two-Parameter Interaction

Table 4 reports the two-parameter interaction between $N$ and $\alpha_{\text{SC}}$, the two strongest Sobol drivers. The interaction between $N$ and $\alpha_{\text{SC}}$ is sub-additive (interaction term $-0.050$); details are in the table notes.

## IA.3 Robustness Across Alternative Specifications

Table 5 reports $\rho^*$ under alternative functional forms for the aggregation mapping $g_1$ and cost specification.[13]

[13] The footnote to Assumption A2 in Section 4 claims the safety illusion survives under concave and convex alternatives to the linear $g_1$. Table 5 provides formal evidence.

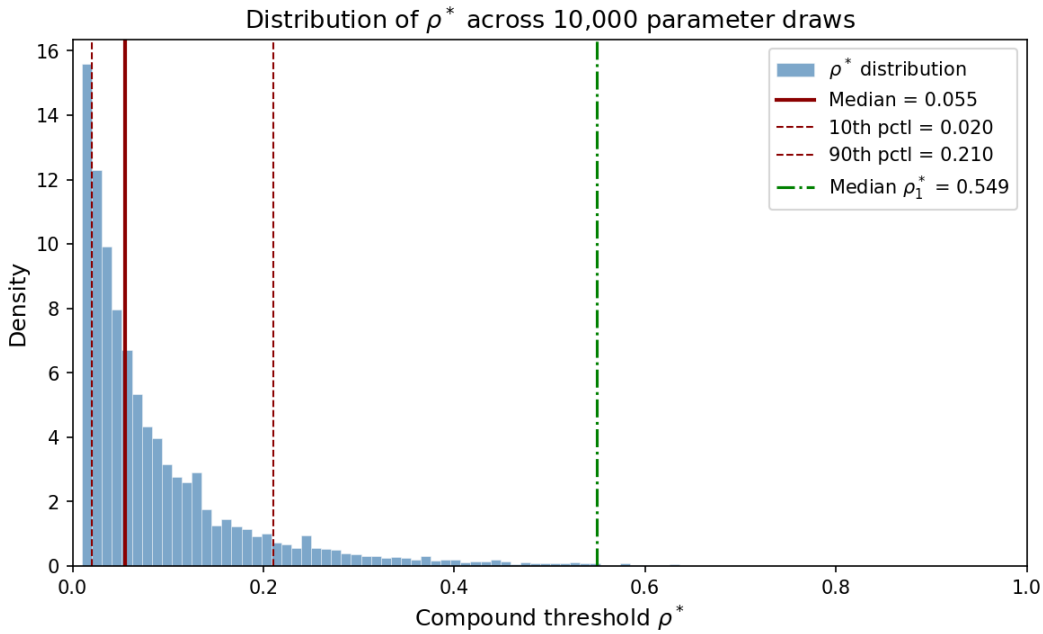


Figure 2: Distribution of the compound fragility threshold $\rho^*$ across 10,000 Monte Carlo draws from uniform parameter ranges in Table 1. Vertical line: baseline $\rho^* = 0.215$. Median: 0.055; [p10, p90] = [0.020, 0.210]. The safety illusion ($\rho^* < \rho_1^*$) holds in 100% of draws.

## IA.4 Monte Carlo Histogram

## IA.5 First-Order Sobol Sensitivity Indices

Table 6 reports first-order Sobol sensitivity indices for all eleven free model parameters, computed from the 10,000-draw Monte Carlo exercise. The index for each parameter is approximated by the $R^2$ of a regression of $\rho^*$ on that parameter in isolation, using Latin Hypercube sampling across the full ranges in Table 1.

## IA.6 Limit-Case Verification and Specification Checks

**DSZ limit.** At $\rho = 1$, the model nests Danielsson et al. [2012] as a limit case: $N_{\text{eff}} = 1.000$ (single effective market maker), $\rho_{\text{eff}} = 1.000$ (perfect correlation), $\theta^* = 0.950$ (maximal crisis threshold), and $\mu_I \approx 0$ (private information acquisition ceases). Net market-making profit is negative ($-11.505$), confirming the no-equilibrium result under quadratic costs.

**Signal-level vs. withdrawal-level correlation.** The baseline specification uses $\rho$ (signal-level correlation) in $g_3$, while an alternative uses $\rho_W$ (withdrawal-level correlation from the Bernoulli correction). Under the $\rho_W$ specification, $\rho^* = 0.380$ (versus 0.215 at baseline), and the safety illusion width narrows to 0.291. The baseline is the conservative choice: because

Table 6: First-order Sobol sensitivity indices for $\rho^*$

| Parameter | Description | Sobol Index | Rank |
|---|---|---|---|
| $\alpha_{\text{SC}}$ | Coordination motive | 0.214 | 1 |
| $N$ | AI-equipped dealers | 0.160 | 2 |
| $c_P$ | Private signal cost | 0.102 | 3 |
| $\sigma$ | Signal noise (std dev) | 0.007 | 4 |
| $k_{\text{GY}}$ | GY complementarity | 0.000 | 5 |
| $C$ | Cost coefficient | 0.000 | 6 |
| $\tilde{m}$ | Feedback gain | 0.000 | 7 |
| $\gamma$ | Risk aversion | 0.000 | 8 |
| $p_w$ | Withdrawal probability | 0.000 | 9 |
| $\pi_0$ | Outside option | 0.000 | 10 |
| $D$ | Revenue coefficient | 0.000 | 11 |

*Notes:* Sobol indices estimated as $R^2$ from a univariate regression of $\rho^*$ on each parameter across 10,000 Latin Hypercube draws. The top three drivers—$\alpha_{\text{SC}}$, $N$, and $c_P$—are precisely those parameters with direct empirical anchors (Goldstein-Pauzner 2005, FRBNY dealer counts, and cost-of-information estimates). The information acquisition elasticity $|a|$ does not appear as a free parameter because it is fully determined by the Grossman-Stiglitz indifference condition (see Table 1 notes); variation in $|a|$ across draws is induced by variation in $(c_P, k_P, k_A, \alpha, \beta)$. The withdrawal probability $p_w$ has a near-zero Sobol index in the baseline specification because $g_3$ uses signal-level $\rho$ rather than the Bernoulli withdrawal correlation $\rho_W$; under the $\rho_W$ specification (Table IA.3), $p_w$'s index rises to approximately 8%.

$\rho_W < \rho$ for tail events ($p_w = 0.05$), using $\rho_W$ in $g_3$ raises $N_{\text{eff}}$ and weakens the amplification loop, making the bifurcation harder to trigger. The safety illusion result holds under both specifications.